Liste. Stive. Cozi

SD 2021/2022

Conținut

Tipurile abstracte LLin, LLinOrd, Stiva, Coada
- Liste liniare
- Implementarea cu tablouri
- Implementarea cu liste simplu înlănțuite
- Liste liniare ordonate
- Stive
- Cozi

Aplicație la conversie de expresii aritmetice

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Aplicație la conversie de expresii aritmetice

Liste liniare – exemple

- Studenți
  - *(Adriana, George, Luiza, Maria, Daniel)*
- Examene
  - *(Mate, Logică, SD, ACSO, IP, ENG)*
- Zilele săptămânii
  - *(L, M, Mi, J, V, S, D)*
- Lunile anului
  - *(Ian, Feb, Mar, Apr, Mai, Iun, Iul, Aug, Sep, Oct, Nov, Dec)*

Tipul abstract `LLin`

- OBIECTE: $L = (e_0, \cdots, e_{n-1}), \quad n \geq 0$
- $e_i \in$ `Elt` (tipul abstract al elementelor)
- Relații:
  - e_0 primul element al listei;
  - e_{n-1} ultimul element al listei;
  - e_i elementul predecesor lui e_{i+1}.

LLin – operații

- `listaVida()`
  - intrare: nimic
  - ieșire: $L = ()$ (lista cu zero elemente)

- `insereaza()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad k \in \texttt{Nat}, \quad e \in \texttt{Elt}$
  - ieșire:
    - $L = (\cdots, e_{k-1}, e, e_k, \cdots)$, dacă $0 \leq k \leq n$
    - eroare în caz contrar

insereaza() – exemple

$$L = (a, b, c, d, e, f, g)$$

- insereaza$(L, 0, x)$ $\Rightarrow$ $L = (x, a, b, c, d, e, f, g)$

  Obs. indexul elementelor $a, \cdots, g$ crește cu 1.

- insereaza$(L, 2, x)$ $\Rightarrow$ $L = (a, b, x, c, d, e, f, g)$

- insereaza$(L, 7, x)$ $\Rightarrow$ $L = (a, b, c, d, e, f, g, x)$

- insereaza$(L, 10, x)$ $\Rightarrow$ eroare

- insereaza$(L, -7, x)$ $\Rightarrow$ eroare

LLin – operații

- `elimina()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad k \in \texttt{Nat}$
  - ieșire:
    - $L = (\cdots, e_{k-1}, e_{k+1}, \cdots)$, dacă $0 \leq k \leq n-1$
    - eroare în caz contrar

LLin – operații

- `elimina()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad k \in \texttt{Nat}$
  - ieșire:
    - $L = (\cdots, e_{k-1}, e_{k+1}, \cdots)$, dacă $0 \leq k \leq n-1$
    - eroare în caz contrar

Exemple:

$$L = (a, b, c, d, e, f, g)$$

- $\texttt{elimina}(L, 2) \quad \Rightarrow \quad L = (a, b, d, e, f, g)$
  Obs. indexul elementelor $d, \cdots, g$ descrește cu 1.
- $\texttt{elimina}(L, 10) \quad \Rightarrow \quad$ eroare
- $\texttt{elimina}(L, -7) \quad \Rightarrow \quad$ eroare

LLin – operații

- `alKlea()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad k \in \texttt{Nat}$
  - ieșire:
    - e_k, dacă $0 \leq k \leq n-1$
    - eroare în caz contrar

LLin – operații

- `alKlea()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad k \in \texttt{Nat}$
  - ieșire:
    - e_k, dacă $0 \leq k \leq n-1$
    - eroare în caz contrar

Exemple:

$$L = (a, b, c, d, e, f, g)$$

- $\texttt{alKlea}(L, 0) \quad \Rightarrow \quad a$
- $\texttt{alKlea}(L, 2) \quad \Rightarrow \quad c$
- $\texttt{alKlea}(L, 6) \quad \Rightarrow \quad g$
- $\texttt{alKlea}(L, 20) \quad \Rightarrow \quad$ eroare
- $\texttt{alKlea}(L, -2) \quad \Rightarrow \quad$ eroare

LLin – operații

- `elimTotE()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad e \in \texttt{Elt}$
  - ieșire:
    - lista L din care s-au eliminat toate elementele egale cu e

LLin – operații

- `elimTotE()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad e \in \texttt{Elt}$
  - ieșire:
    - lista L din care s-au eliminat toate elementele egale cu e

---

Exemple:

$$L = (a, b, c, a, b, c, a)$$

- $\texttt{elimTotE}(L, a) \quad \Rightarrow \quad (b, c, b, c)$
- $\texttt{elimTotE}(L, c) \quad \Rightarrow \quad (a, b, a, b, a)$
- $\texttt{elimTotE}(L, d) \quad \Rightarrow \quad (a, b, c, a, b, c, a)$

LLin – operații

- `parcurge()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1})$, o procedură / funcție `viziteaza()`
  - ieșire:
    - lista L în care toate elementele au fost procesate aplicând `viziteaza()`

LLin – operații

- `parcurge()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1})$, o procedură / funcție `viziteaza()`
  - ieșire:
    - lista L în care toate elementele au fost procesate aplicând `viziteaza()`

Exemple:

$$L = (1, 2, 3, 1, 2, 3)$$

- `parcurge`$(L,$ `oriDoi()`$)$ $\Rightarrow$ $(2, 4, 6, 2, 4, 6)$
- `parcurge`$(L,$ `incrementeaza()`$)$ $\Rightarrow$ $(2, 3, 4, 2, 3, 4)$

LLin – operații

- `poz()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad e \in \texttt{Elt},$
  - ieșire:
    - prima poziție pe care apare e în L sau
    - -1 dacă e nu apare în L.

LLin – operații

- poz()
  - intrare:
    - $L = (e_0, \cdots, e_{n-1}), \quad e \in \texttt{Elt},$
  - ieșire:
    - prima poziție pe care apare e în L sau
    - -1 dacă e nu apare în L.

Exemple:

$$L = (a, b, c, a, b, c, d)$$

- $\text{poz}(L, a) \quad \Rightarrow \quad 0$
- $\text{poz}(L, c) \quad \Rightarrow \quad 2$
- $\text{poz}(L, d) \quad \Rightarrow \quad 6$
- $\text{poz}(L, x) \quad \Rightarrow \quad -1$

LLin – operații

- `lung()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1})$,
  - ieșire:
    - n – lungimea listei L.

LLin – operații

- `lung()`
  - intrare:
    - $L = (e_0, \cdots, e_{n-1})$,
  - ieșire:
    - n – lungimea listei L.

Exemple:

$$L = (a, b, c, a, b, c, d)$$

- $\texttt{lung}(L) \quad \Rightarrow \quad 7$

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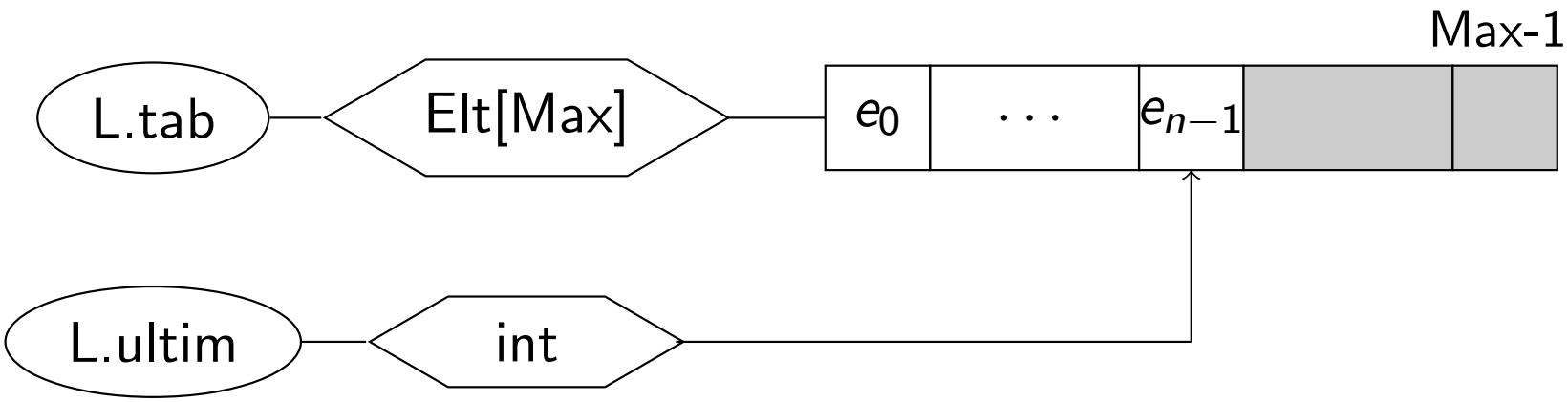
Aplicație la conversie de expresii aritmetice

LLin – implementarea cu tablouri

- Reprezentarea obiectelor $L = (e_0, \cdots, e_{n-1})$



- L este o *structură*
  - *L.tab* – un câmp de tip tablou pentru memorarea elementelor;
  - *L.ultim* – un câmp pentru memorarea poziției ultimului element.

LLin – implementarea cu tablouri

- `insereaza()`
  - deplasează elementele de pe pozițiile $k, k+1, \ldots$ la dreapta cu o poziție;
  - inserează e pe poziția k;
  - excepții:
    - $k < 0, \quad k > L.ultim + 1 \quad (n)$
    - $L.ultim = Max - 1.$

LLin – implementarea cu tablouri

procedure *insereaza*(L, k, e)
begin
  if $(k < 0$ `or` $k > L.ultim + 1)$ **then**
    throw "eroare-pozitie incorecta"
  if $(L.ultim >= Max - 1)$ **then**
    throw "eroare-spatiu insuficient"
  for $j \leftarrow L.ultim$ downto k **do**
    $L.tab[j + 1] \leftarrow L.tab[j]$
  $L.tab[k] \leftarrow e$
  $L.ultim \leftarrow L.ultim + 1$
end

- Timpul de execuție: $O(n)$.

LLin – implementarea cu tablouri

- `parcurge()`

procedure *parcurge(L, viziteaza())*
begin
    for $i \leftarrow 0$ to $L.ultim$ **do**
        viziteaza($L.tab[i]$)
end

- Dacă *viziteaza*() procesează un element în $O(1)$, atunci *parcurge*() procesează lista în $O(n)$ (n este numărul elementelor listei).

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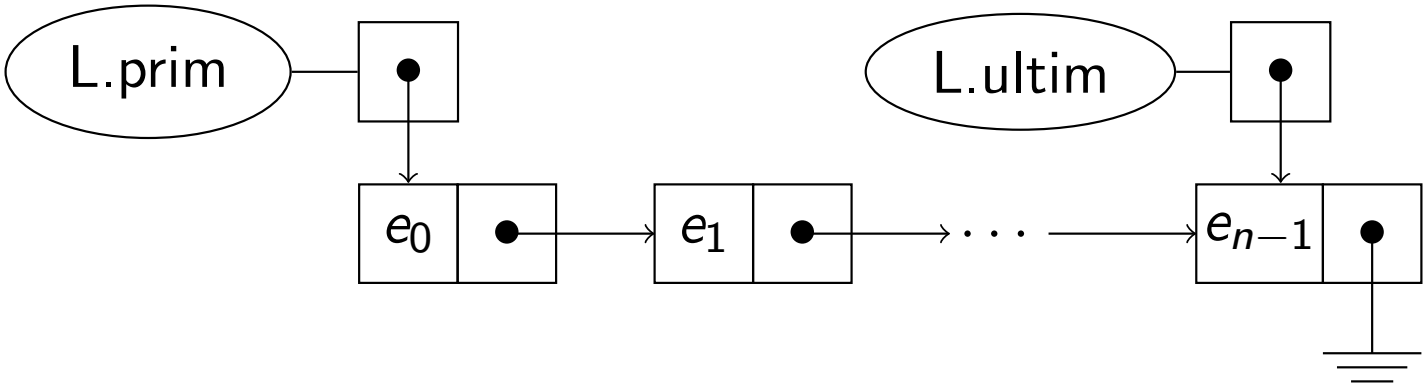
Aplicație la conversie de expresii aritmetice

LLin – implementarea cu structuri înlănțuite

- Reprezentarea obiectelor $L = (e_0, \cdots, e_{n-1})$



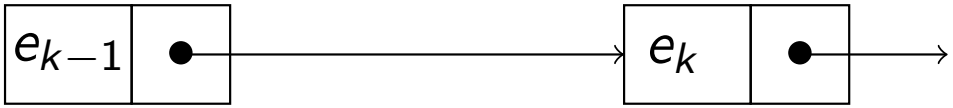
- L este o *structură* cu două câmpuri
  - $L.prim$ – pointer la primul element al listei;
  - $L.ultim$ – pointer la ultimul element al listei.
- un nod $*$ p (aflat la adresa din p) are două câmpuri:
  - $p->elt(=e_i)$ – memorează informația utilă din nod;
  - $p->succ$ – memorează adresa nodului succesor.

LLin – implementarea cu structuri înlănțuite

- `insereaza()`
  - parcurge elementele de pe pozițiile $0, 1, \ldots, k-1$;
  - inserează un nou element după poziția $k-1$;
    - creează noul nod;
    - memorează informații;
    - reface legături.
  - excepții:
    - lista vidă;
    - $k = 0$;
    - $k = n$;
    - $k < 0, k > n$.

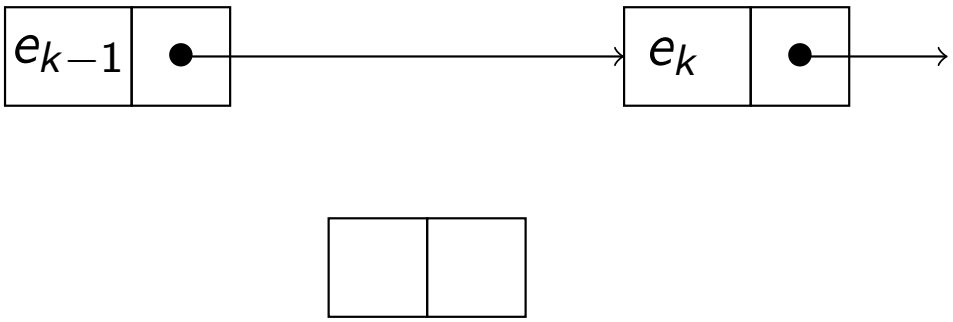
LLin – implementarea cu structuri înlănțuite

- Cazul general



LLin – implementarea cu structuri înlănțuite

- Cazul general



LLin – implementarea cu structuri înlănțuite

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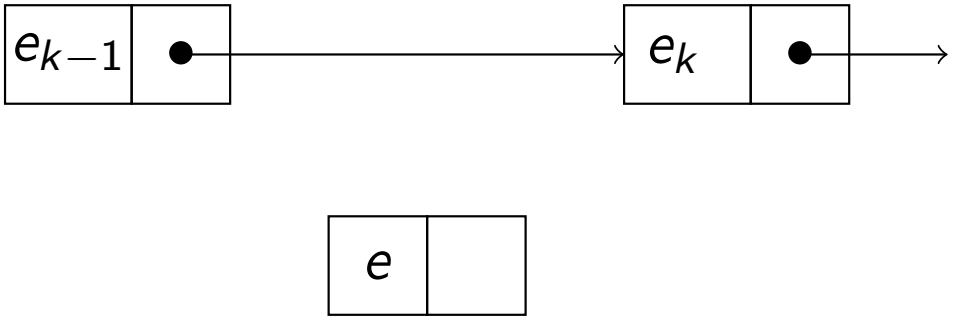

LLin – implementarea cu structuri înlănțuite

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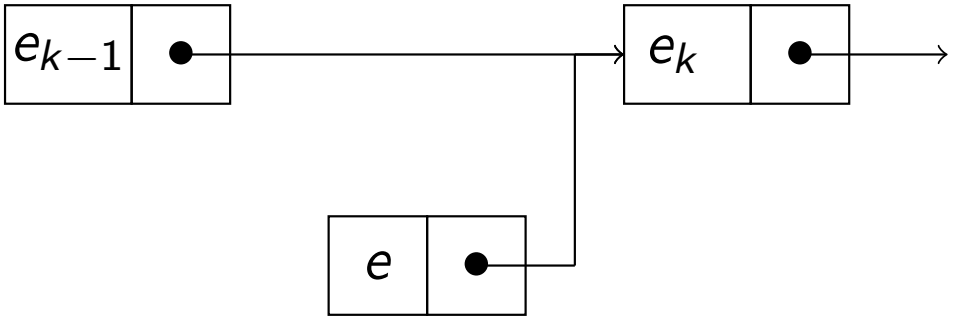

LLin – implementarea cu structuri înlănțuite

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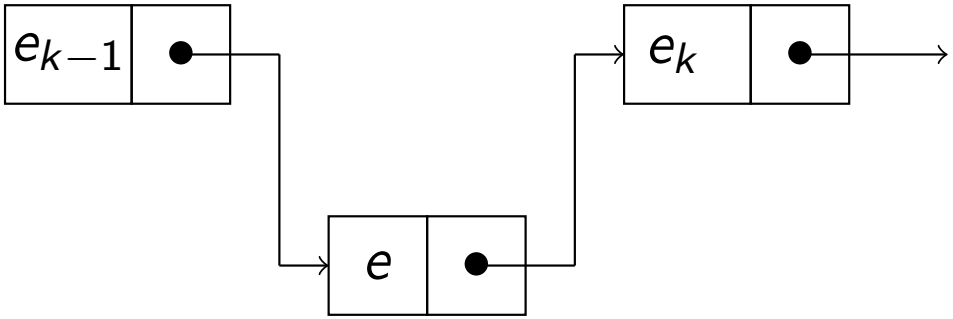
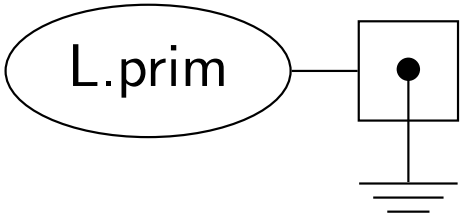
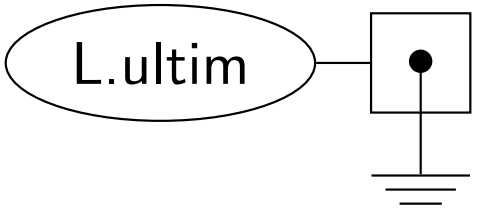


LLin – implementarea cu structuri înlănțuite

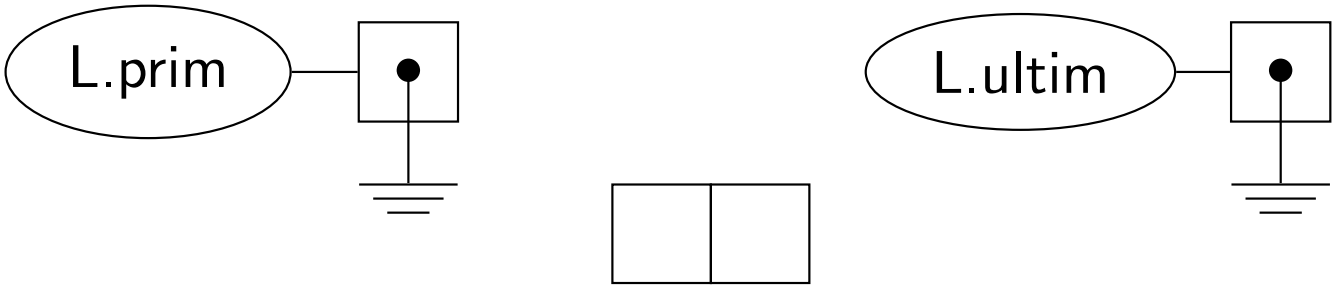
- Cazul particular: lista vidă





LLin – implementarea cu structuri înlănțuite

- Cazul particular: lista vidă



LLin – implementarea cu structuri înlănțuite

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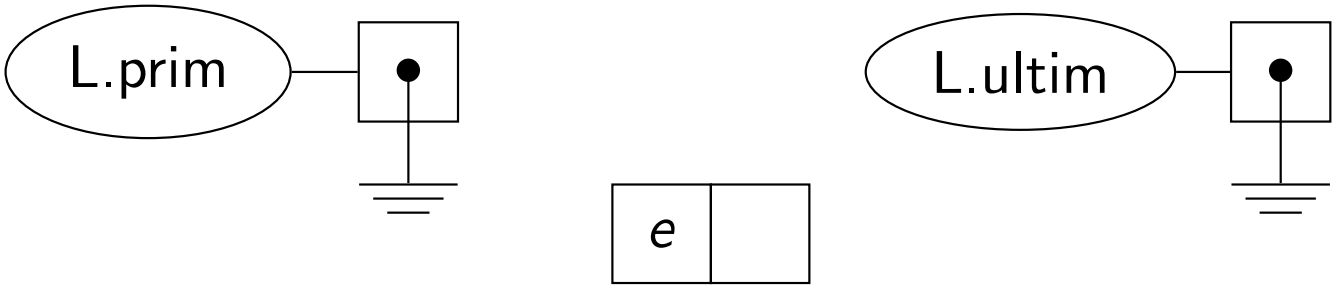

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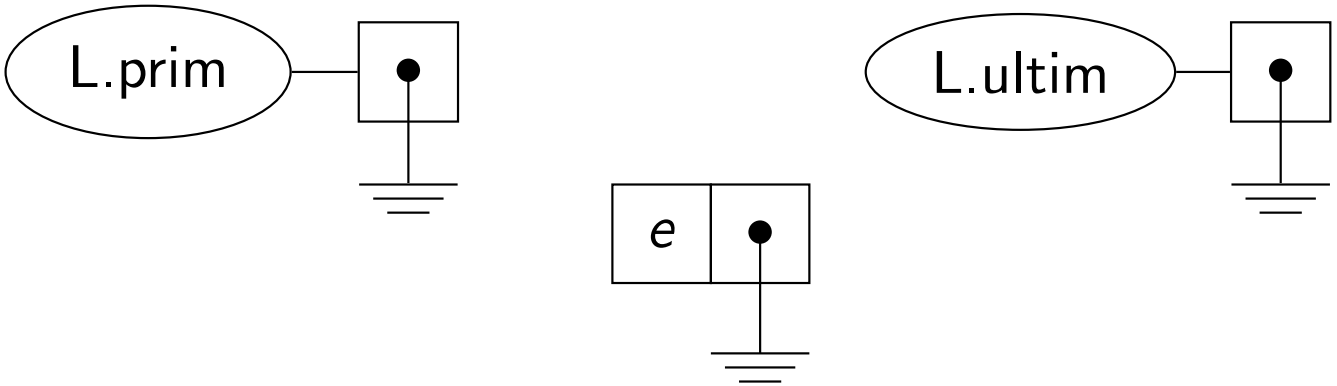
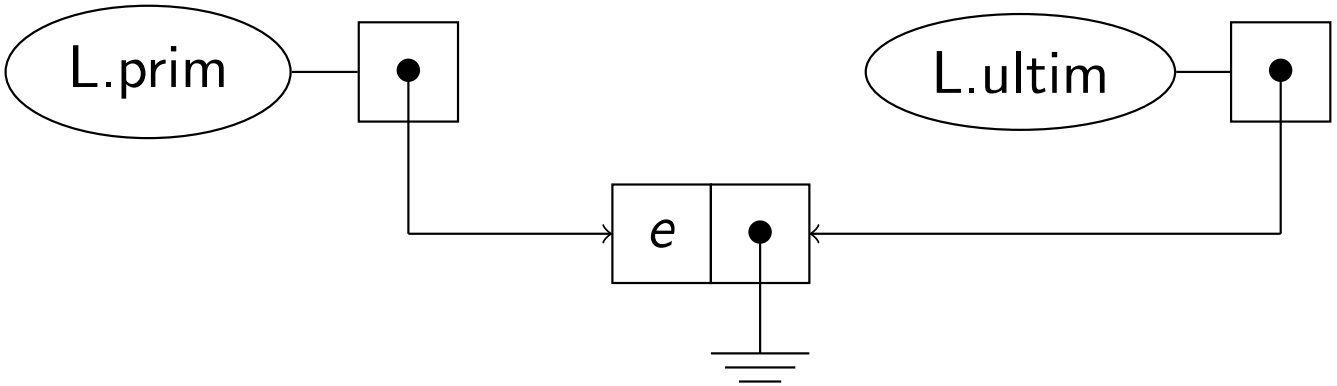


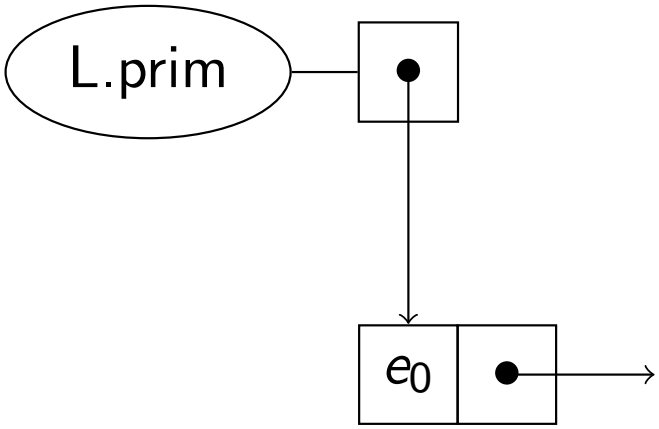
LLin – implementarea cu structuri înlănțuite

- Cazul particular: lista vidă



LLin – implementarea cu structuri înlănțuite

- Cazul particular: inserarea la începutul listei



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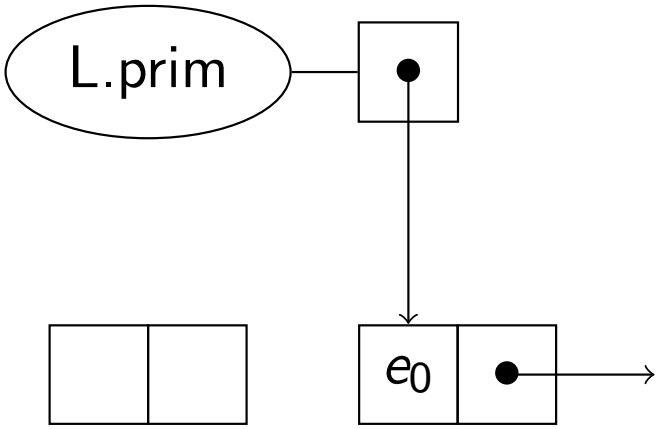

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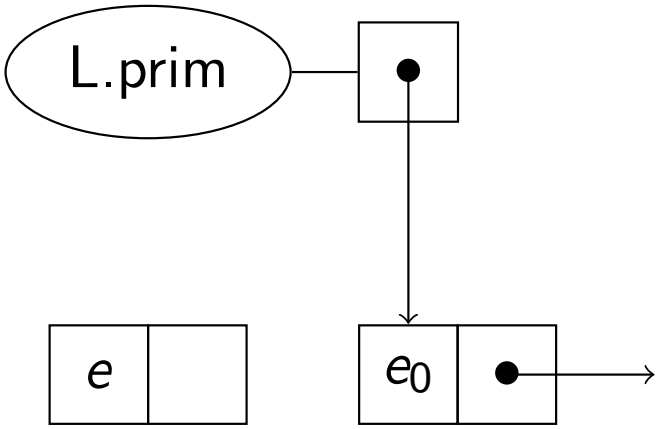

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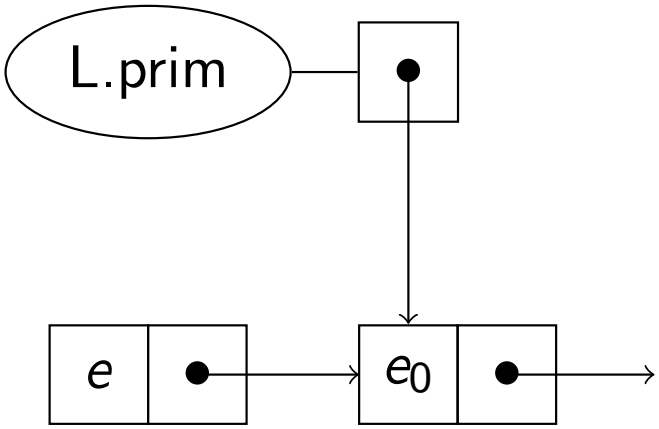
- Cazul particular: inserarea la începutul listei



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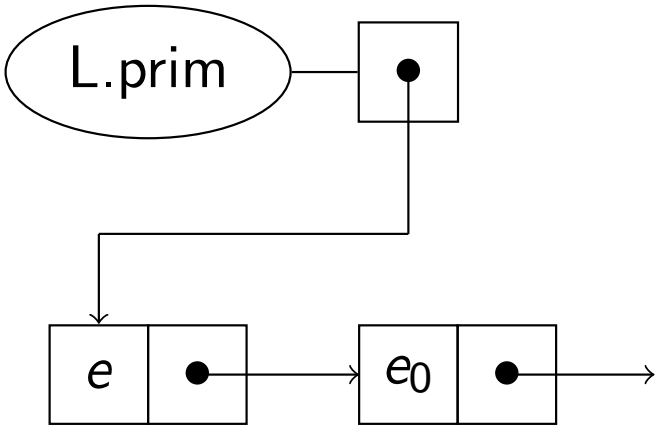

LLin – implementarea cu structuri înlănțuite

```
procedure insereaza(L, k, e)
begin
    if (k < 0) or (k! = 0 and L.prim == NULL) then
        throw "eroare-pozitie incorecta"
    new(q);    q->elt ← e
    if (k == 0 or L.prim == NULL) then
        q->succ ← L.prim;    L.prim ← q
        if (L.ultim == NULL) then
            L.ultim ← q
    else
        p ← L.prim;    j ← 0
        while (j < k − 1 and p! = L.ultim) do
            p ← p->succ;    j ← j + 1
        if (j < k − 1) then
            throw "eroare-pozitie incorecta"
        q->succ ← p->succ;    p->succ ← q
        if (p == L.ultim) then
            L.ultim ← q
end
```

LLin – aplicație

- Linie poligonală de puncte.
  - Punct: structură cu două câmpuri x și y;
  - crearea unei liste

```
procedure creeazaLista(L)
begin
    L ← listaVida()
    /* citeste n */
    for i ← 0 to n − 1 do
        /* citeste p.x, p.y */
        insereaza(L, 0, p)
end
```

- Obs. Timpul de execuție depinde de implementare.

LLin – aplicație

- Multiplică cu 2 coordonatele unui punct:

procedure *ori2Punct(p)*
begin
    $p.x \leftarrow p.x * 2$
    $p.y \leftarrow p.y * 2$
end

- Multiplică cu 2 coordonatele unei linii poligonale:

procedure *ori2Linie(p)*
begin
    $parcurge(L, ori2Punct())$
end

LLin – aplicație

- translatează punct:

procedure *trPunct*(p, dx, dy)
begin
    $p.x \leftarrow p.x + dx$
    $p.y \leftarrow p.y + dy$
end

- translatează linie poligonală:

procedure *trLinie*(L, dx, dy)
begin
    parcurge$(L, trPunct())$
end

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Aplicație la conversie de expresii aritmetice

Liste liniare ordonate: `LLinOrd`

- OBIECTE:
  $L = (e_0, \cdots, e_{n-1}), \quad n \geq 0, \quad e_i \in \texttt{Elt}, \quad e_0 \leq e_1 \leq \cdots \leq e_{n-1}$

- Operații:
  - `listaVida()`
    - intrare: nimic
    - ieșire: $L = ()$ (lista cu zero elemente)
  - `insereaza()`
    - intrare: $L = (e_0, \ldots, e_{n-1}), \quad e \in \texttt{Elt}$
    - ieșire: $L = (\cdots, e_{k-1}, e, e_k, \cdots)$, dacă $e_{k-1} \leq e \leq e_k$
      $(e_{-1} = -\infty, e_n = +\infty)$

Liste liniare ordonate: `LLinOrd`

- `elimina()`
  - intrare: $L = (e_0, \ldots, e_{n-1}), \quad e \in \texttt{Elt}$
  - ieșire: $L = (\cdots, e_{k-1}, e_{k+1}, \cdots)$, dacă $e = e_k$
    eroare în caz contrar
- `alKlea()`
- `parcurge()`
- `poz()`

LLinOrd – implementarea cu tablouri

```
function poz(L, e)
begin
    p ← 0;    q ← L.ultim
    m ← (p + q)/2
    while (L.tab[m]! = e and p < q) do
        if (e < L.tab[m]) then
            q ← m–1
        else
            p ← m + 1
        m ← (p + q)/2
    if (L.tab[m] == e) then
        return m
    else
        return −1
end
```

LLinOrd – complexitatea căutării

- Implementarea cu tablouri: $O(\log_2 n)$;
- Implementarea cu liste înlănțuite: $O(n)$;

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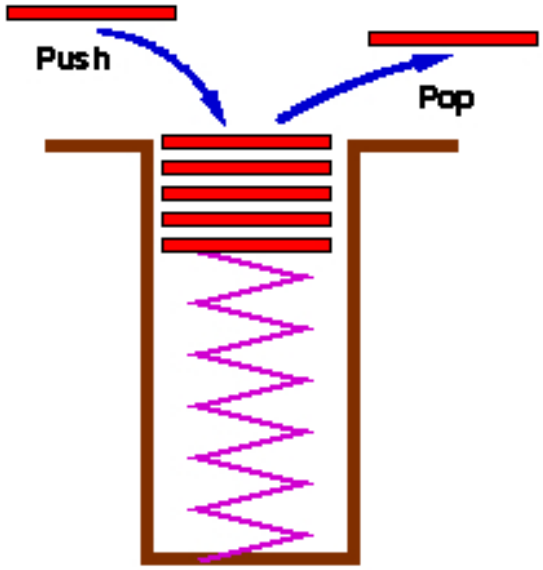
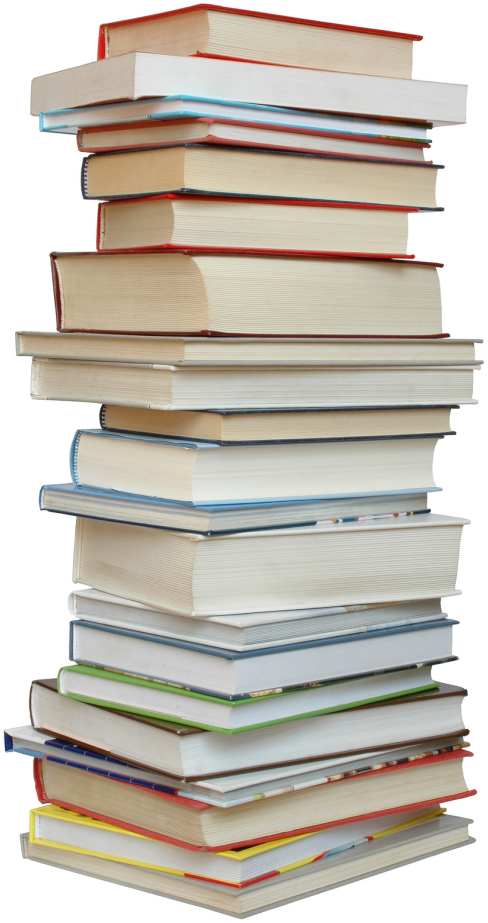
Aplicație la conversie de expresii aritmetice

Stiva





Stiva – aplicații

- Aplicații directe
  - Istoricul paginilor web vizitate într-un browser;
  - Sevența "undo" într-un editor de text;
  - Șiruri de apeluri recursive ale unui subprogram.
- Aplicații indirecte
  - Structură de date auxiliară în anumiți algoritmi;
  - Componentă a altor structuri de date.

Tipul abstract `Stiva`

- OBIECTE:
  Liste în care se cunoaște vechimea elementelor introduse:
  liste LIFO (*Last-In-First-Out*).

- Operații:
  - `stivaVida()`
    - intrare: nimic
    - ieșire: $S = ()$ (lista vidă)
  - `esteVida()`
    - intrare: $S \in$ `Stiva`
    - ieșire:
      – **true** dacă S este vidă;
      – **false** dacă S nu este vidă.

Tipul abstract `Stiva`

- Operații:
  - `push()`
    - intrare: $S \in$ `Stiva`, $e \in$ `Elt`
    - ieșire: S la care s-a adăugat e ca ultim element introdus (cel cu vechimea cea mai mică).
  - `pop()`
    - intrare: $S \in$ `Stiva`
    - ieșire:
      – S din care s-a eliminat ultimul element introdus (cel cu vechimea cea mai mică);
      – eroare dacă S este vidă.
  - `top()`
    - intrare: $S \in$ `Stiva`
    - ieșire:
      – ultimul element introdus în S (cel cu vechimea cea mai mică);
      – eroare dacă S este vidă.

Stiva – implementare cu liste

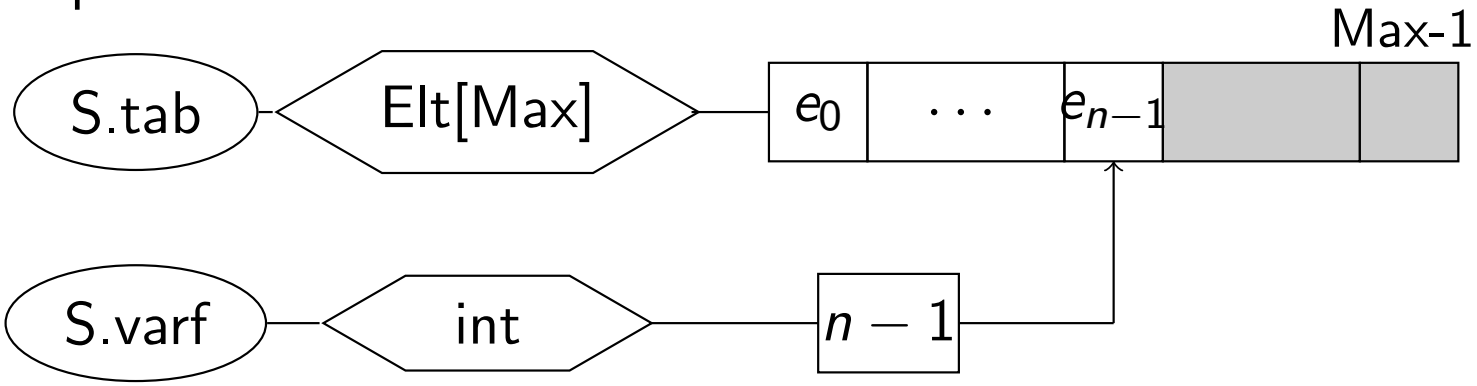
tipul Stiva		tipul LLin
$push(S, e)$	$=$	$insereaza(S, 0, e)$
$pop(S)$	$=$	$elimina(S, 0)$
$top(S)$	$=$	$alKlea(S, 0)$

sau

tipul Stiva		tipul LLin
$push(S, e)$	$=$	$insereaza(S, lung(S), e)$
$pop(S)$	$=$	$elimina(S, lung(S) - 1)$
$top(S)$	$=$	$alKlea(S, lung(S) - 1)$

Stiva – implementarea cu tablouri

- Reprezentarea obiectelor

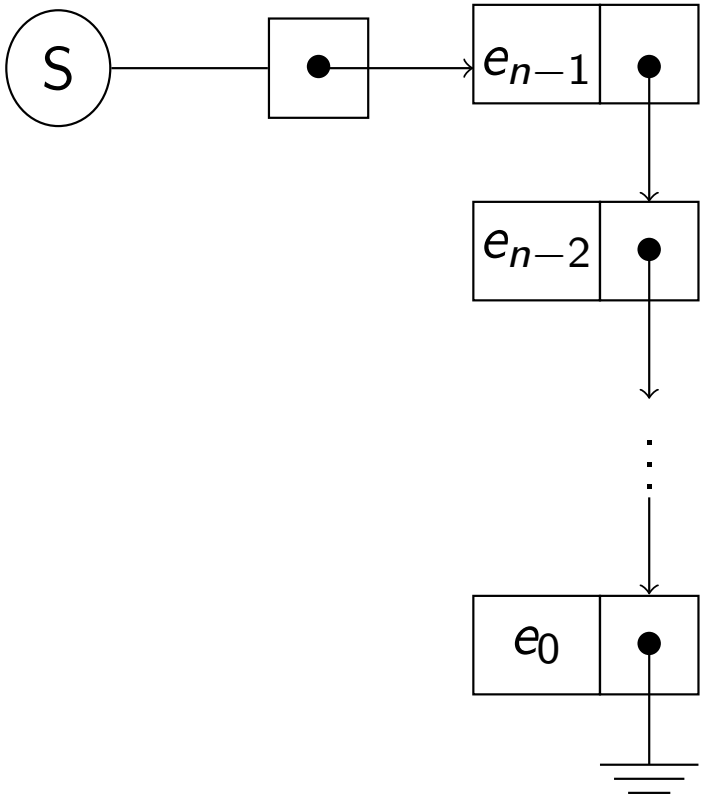


- implementarea operațiilor

procedure *push*(S, e)
begin
  if $S.varf == Max - 1$ **then**
    throw "eroare"
  else
    $S.varf \leftarrow S.varf + 1$
    $S.tab[varf] \leftarrow e$
end

Stiva – implementarea cu structuri înlănțuite

- Reprezentarea obiectelor



Stiva – implementarea cu structuri înlănțuite

- Implementarea operațiilor
  - `push()`

    procedure $push(S, e)$
    begin
    $\quad$ **new**(q)
    $\quad q->elt \leftarrow e$
    $\quad q->succ \leftarrow S$
    $\quad S \leftarrow q$
    end

  - `pop()`

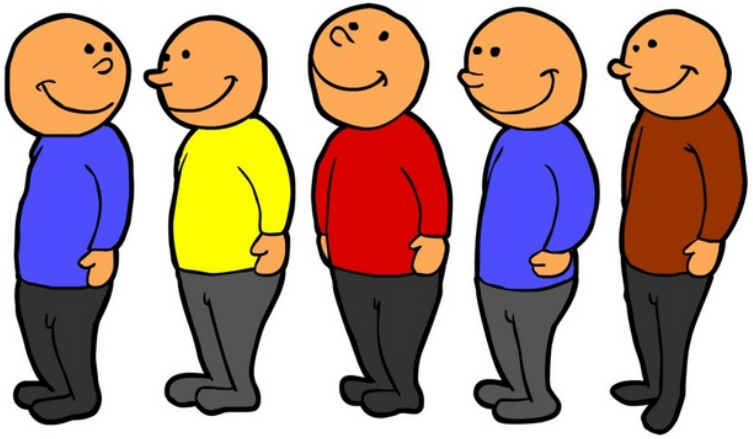
    procedure $pop(S)$
    begin
    $\quad$ **if** $S == NULL$ **then**
    $\quad\quad$ **throw** "eroare"
    $\quad q \leftarrow S$
    $\quad S \leftarrow S->succ$
    delete(q) **end**

Conținut

Tipurile abstracte LLin, LLinOrd, Stiva, Coada
- Liste liniare
- Implementarea cu tablouri
- Implementarea cu liste simplu înlănțuite
- Liste liniare ordonate
- Stive
- Cozi

Aplicație la conversie de expresii aritmetice

Coada



Coada – aplicații

- Aplicații directe
  - Liste / fire de așteptare;
  - Accesul la resurse partajate.
    Exemplu: imprimante.
- Aplicații indirecte
  - Structură de date auxiliară în anumiți algoritmi.

Tipul abstract Coada

- OBIECTE:
  Liste în care se cunoaște vechimea elementelor introduse:
  liste FIFO (*First-In-First-Out*).

- Operații:
  - `coadaVida()`
    - intrare: nimic
    - ieșire: $C = ()$ (lista vidă)
  - `esteVida()`
    - intrare: $C \in$ `Coada`
    - ieșire:
      – **true** dacă C este vidă;
      – **false** dacă C nu este vidă.

Tipul abstract Coada

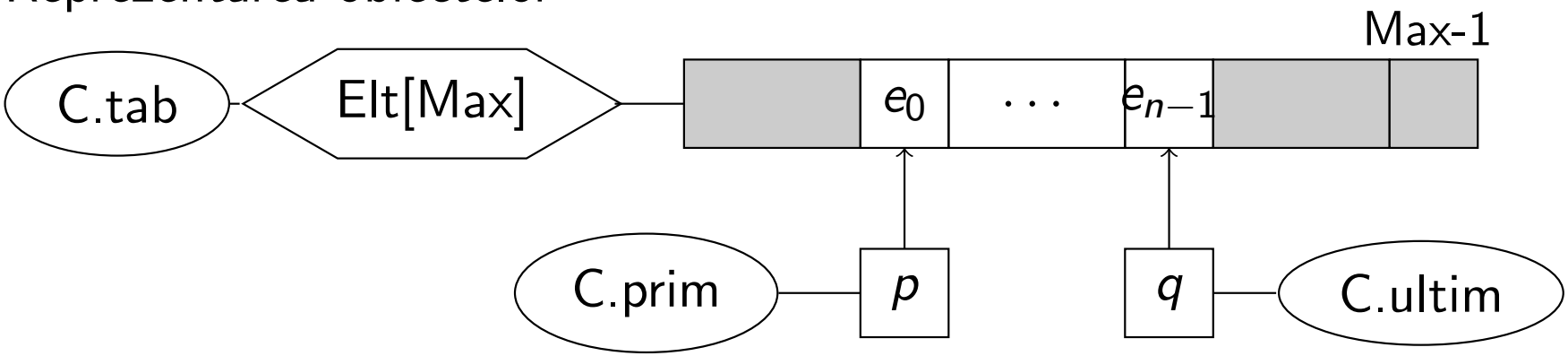
- Operații:
  - `insereaza()`
    - intrare: $C \in$ `Coada`, $e \in$ `Elt`
    - ieșire: C la care s-a adăugat e ca ultim element introdus (cel cu vechimea cea mai mică).
  - `elimina()`
    - intrare: $C \in$ `Coada`
    - ieșire:
      – C din care s-a eliminat primul element introdus (cel cu vechimea cea mai mare);
      – eroare dacă C este vidă.
  - `citeste()`
    - intrare: $C \in$ `Coada`
    - ieșire:
      – primul element introdus în C (cel cu vechimea cea mai mare);
      – eroare dacă C este vidă.

Coada – implementare cu liste

tipul `Coada`		tipul `LLin`
$insereaza(C, e)$	$=$	$insereaza(C, lung(C), e)$
$elimina(C)$	$=$	$elimina(C, 0)$
$citeste(S)$	$=$	$alKlea(C, 0)$

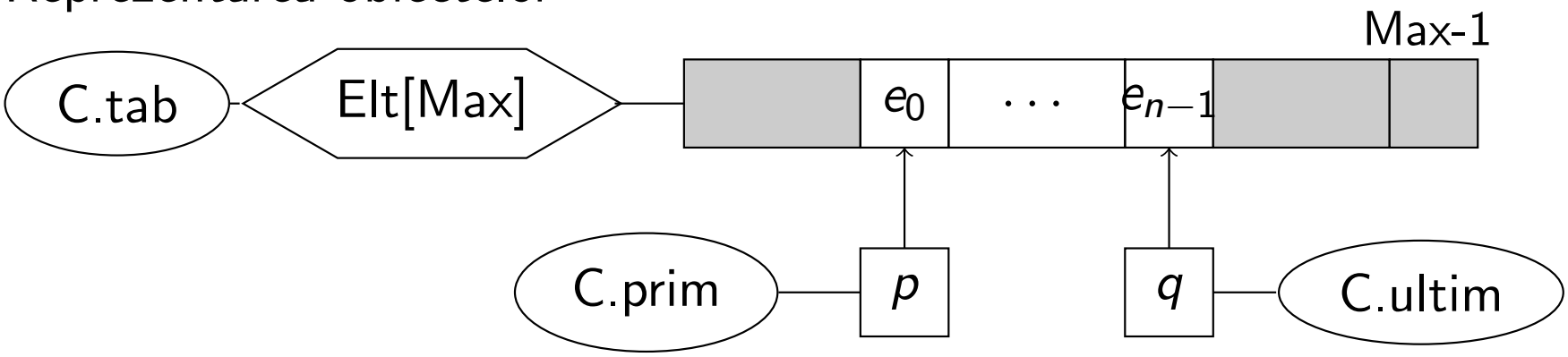
Coada – implementarea cu tablouri

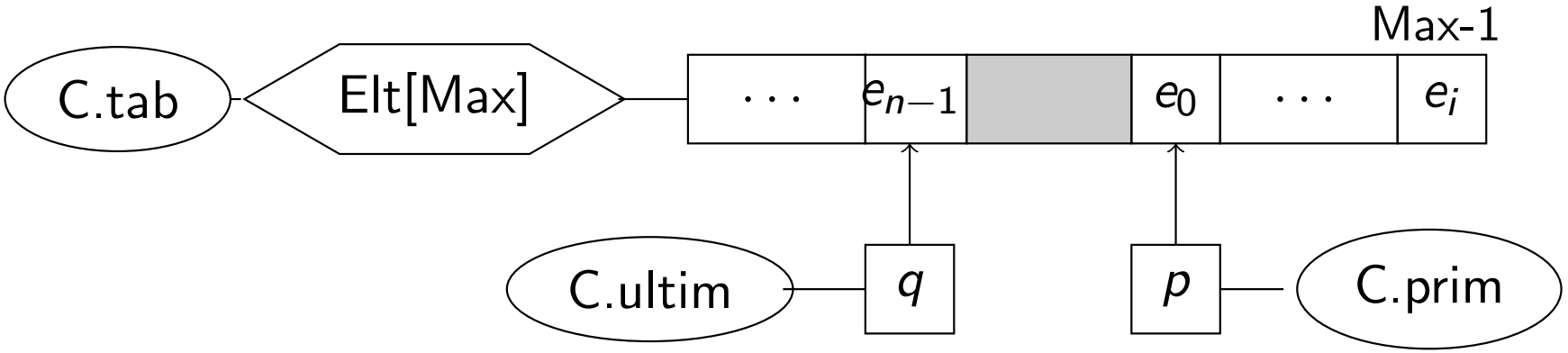
- Reprezentarea obiectelor



Coada – implementarea cu tablouri

- Reprezentarea obiectelor





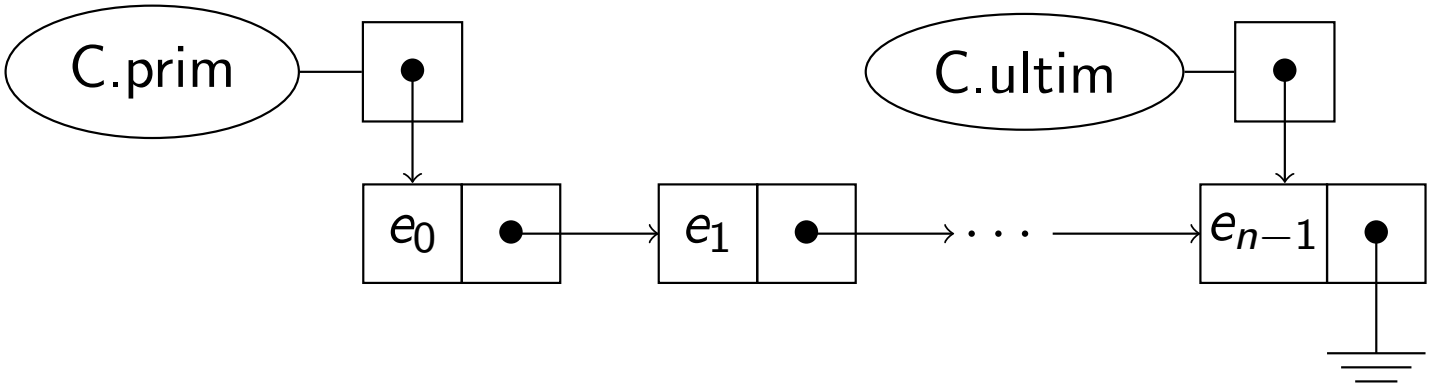
Coada – implementarea cu tablouri

- Implementarea operațiilor
  - `insereaza()`

    procedure *insereaza*(C, e)
    begin
        if $(C.ultim + 1)\%Max == C.prim$ **then**
            throw "eroare"
        else
            $C.ultim \leftarrow (C.ultim + 1)\%Max$
            $C.tab[ultim] \leftarrow e$
    end

Coada – implementarea cu structuri înlănțuite

- Reprezentarea obiectelor



Coada – implementarea cu structuri înlănțuite

- Implementarea operațiilor
  - `insereaza()`

```
procedure insereaza(C, e)
begin
    new(q)
    q->elt ← e
    q->succ ← NULL
    if C.ultim == NULL then
        C.prim ← q
        C.ultim ← q
    else
        C.ultim->succ ← q
        C.ultim ← q
end
```

Conținut

Tipurile abstracte LLin, LLinOrd, Stiva, Coada
- Liste liniare
- Implementarea cu tablouri
- Implementarea cu liste simplu înlănțuite
- Liste liniare ordonate
- Stive
- Cozi

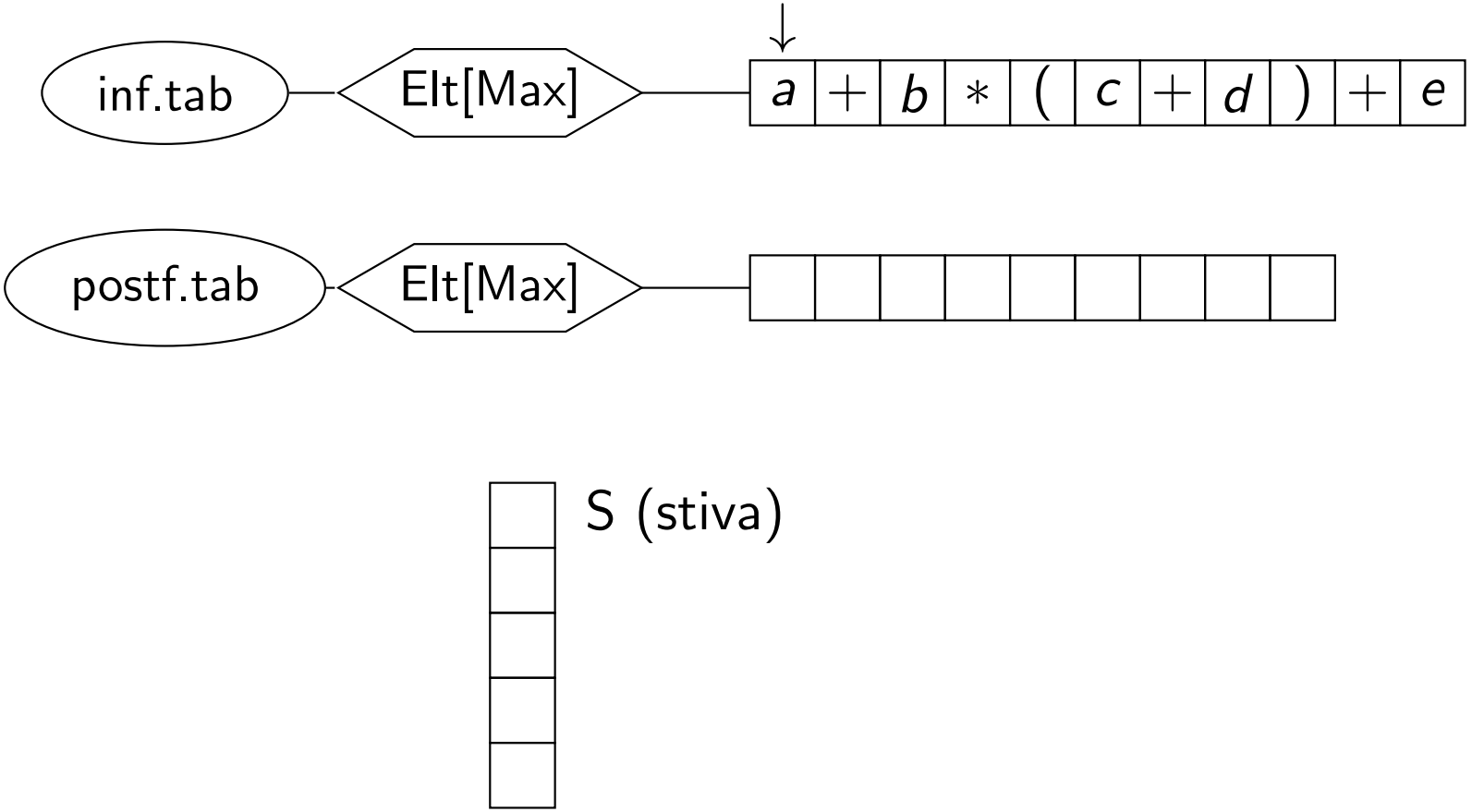
Aplicație la conversie de expresii aritmetice

Aplicație – notația postfixată a expresiilor

- notația infixată
  - $a + b$
  - $a + (b * 2)$
- notația postfixată
  - $a\ b\ +$
  - $a\ b\ 2\ *\ +$
- reguli de precedență
  - $a + b * 2$
- reguli de asociere: $7/3 * 2$
  - la stânga: $(7/3) * 2$
  - la dreapta: $7/(3 * 2)$

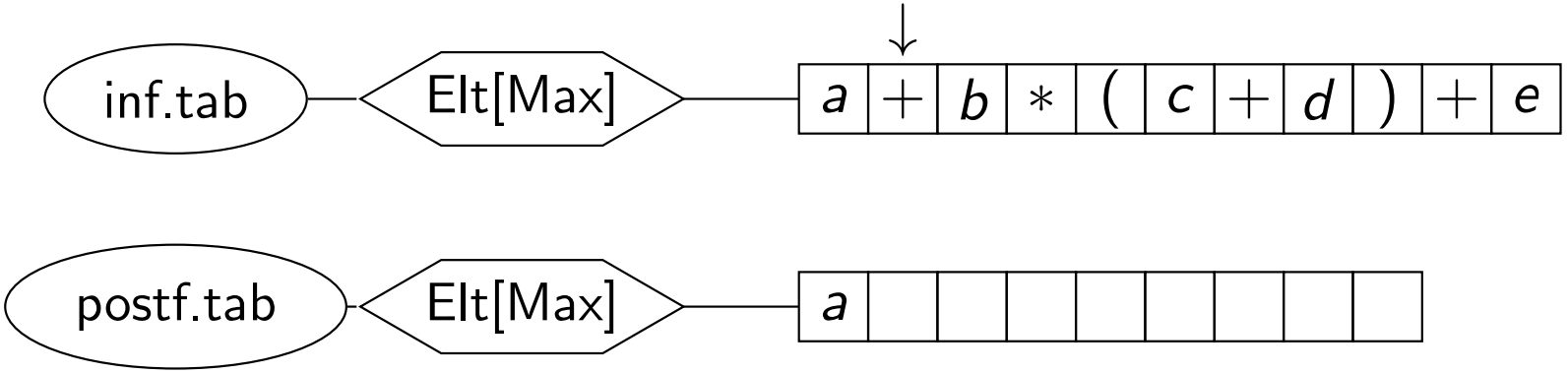
Conversia infixat → postixat

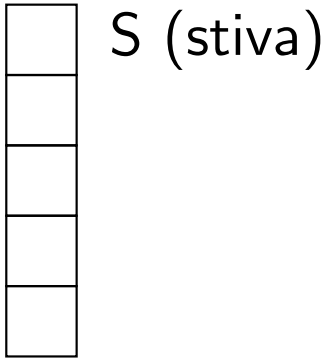
Exemplu: $a + b * (c + d) + e$



Conversia infixat → postixat

Exemplu: $a + b * (c + d) + e$





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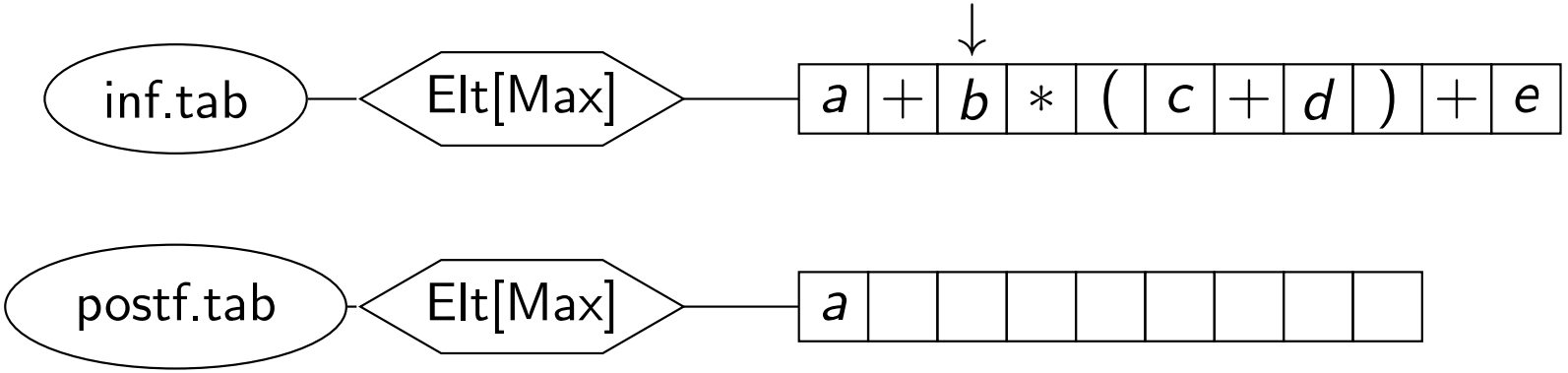
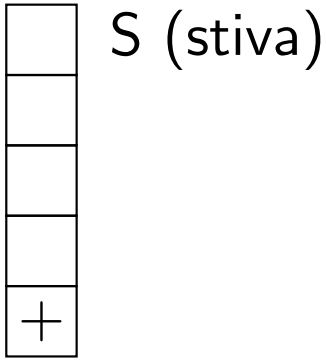
Exemplu: $a + b * (c + d) + e$





Conversia infixat → postixat

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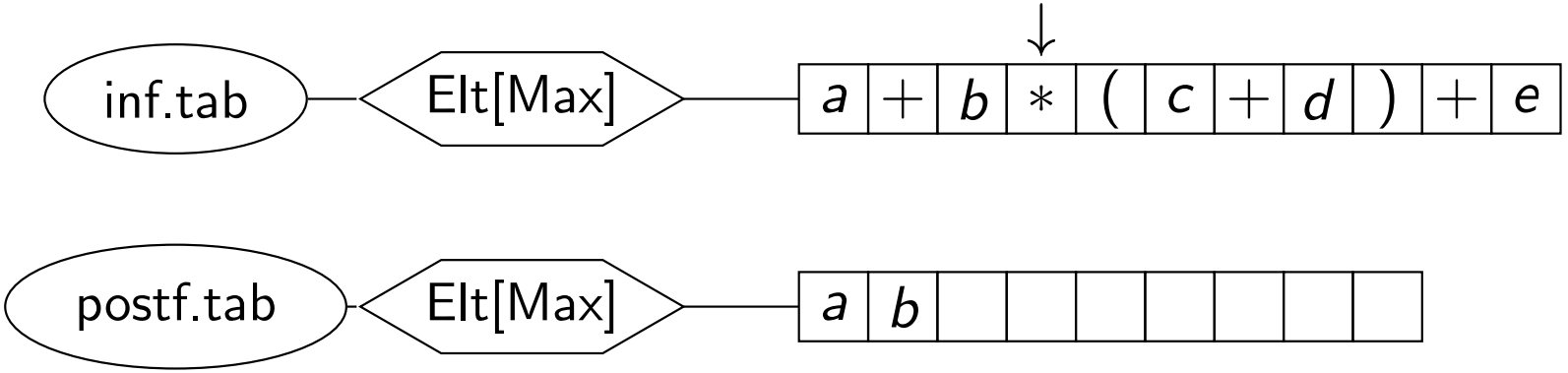

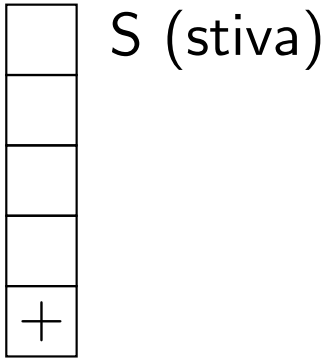

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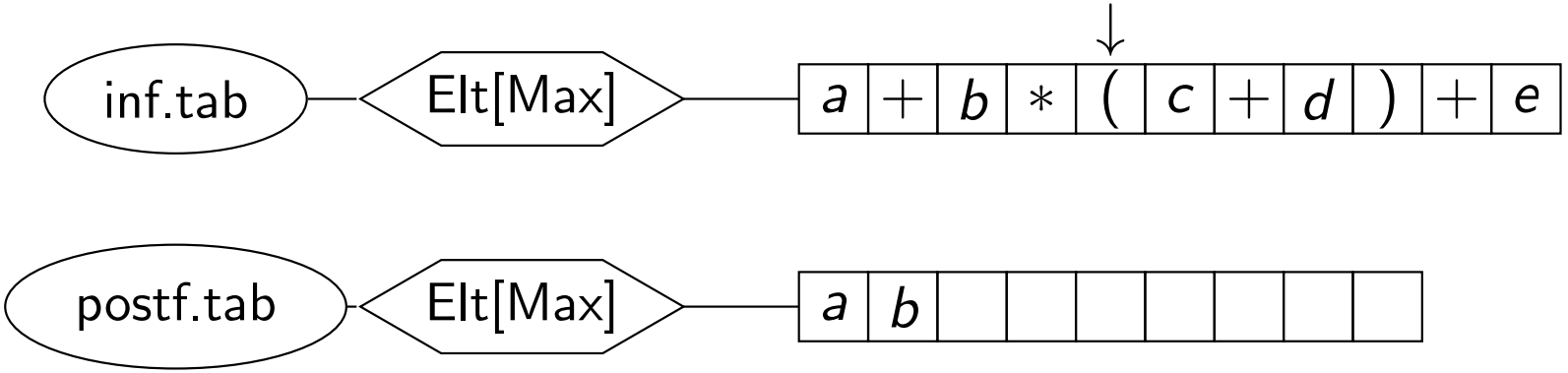
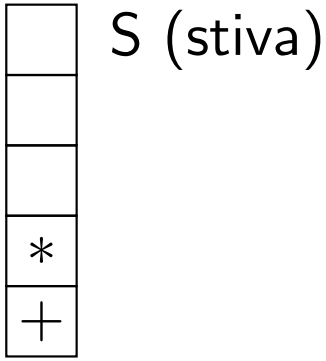
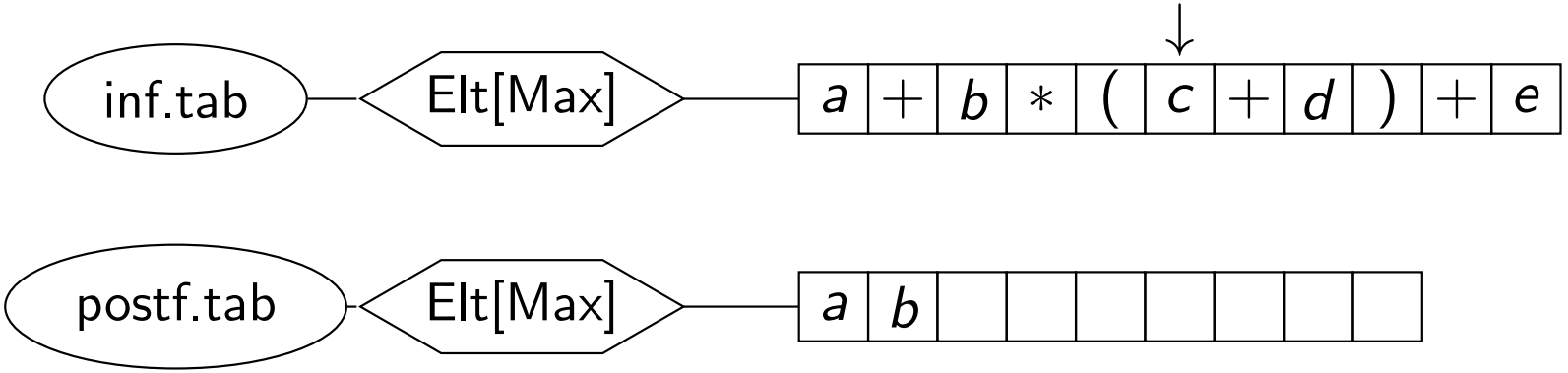
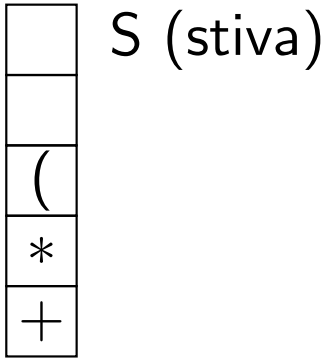




Conversia infixat → postixat

Exemplu: $a + b * (c + d) + e$





Conversia infixat → postixat

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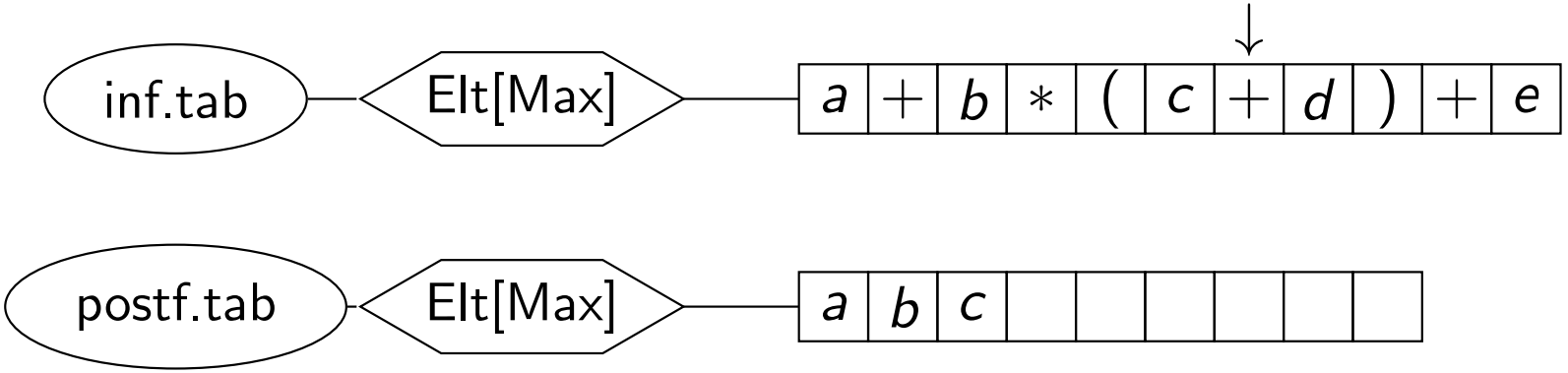

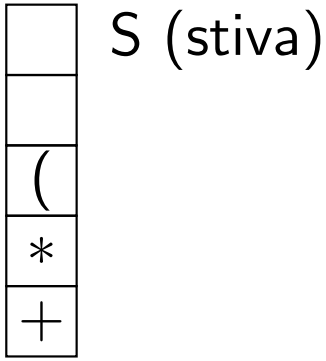

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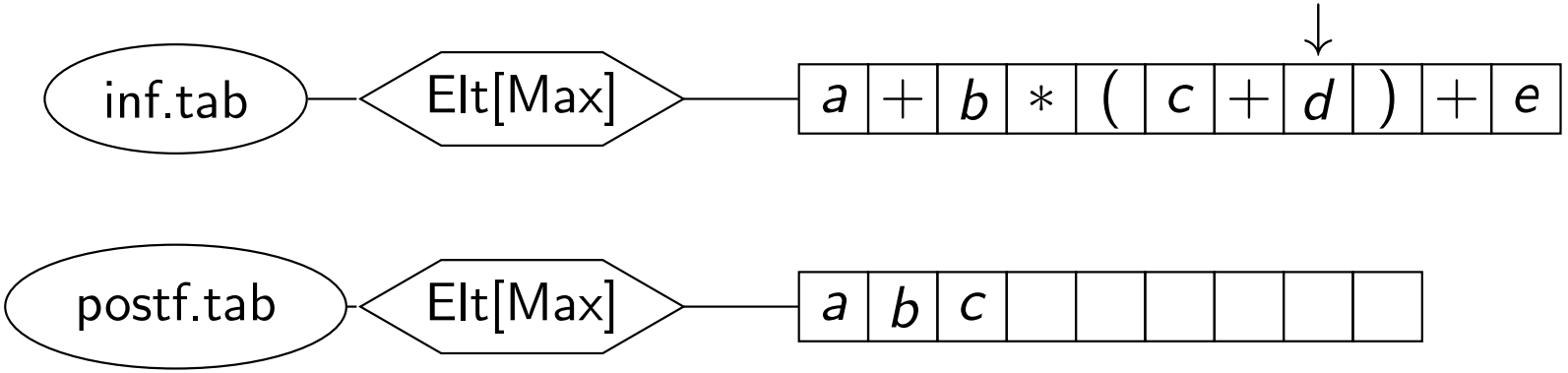

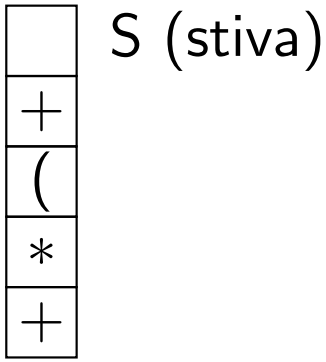

Conversia infixat → postixat

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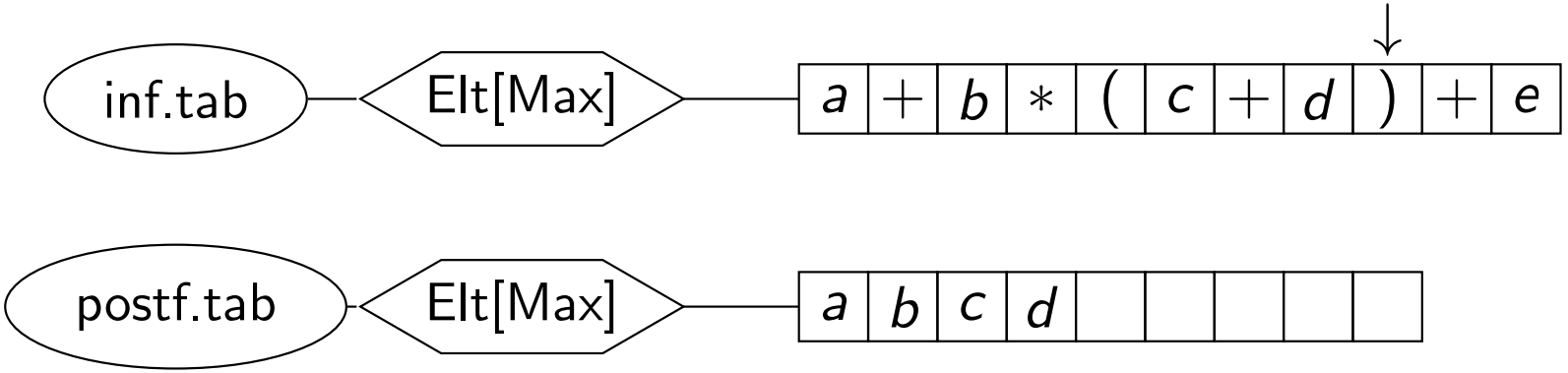
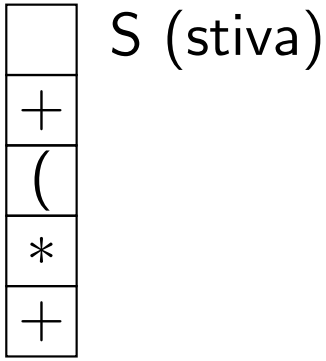
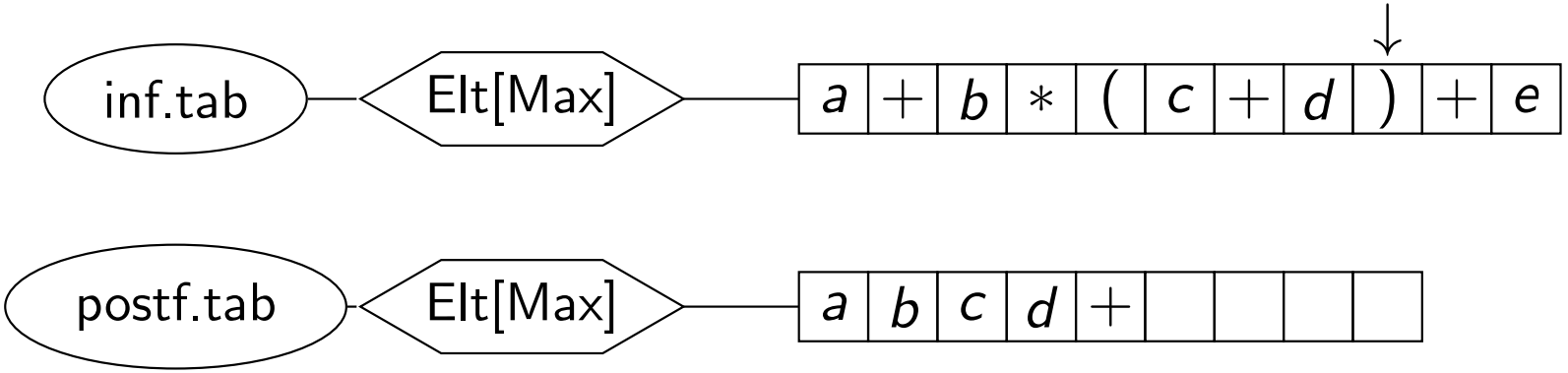
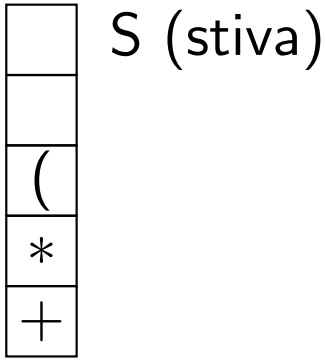




Conversia infixat → postixat

Exemplu: $a + b * (c + d) + e$





Conversia infixat → postixat

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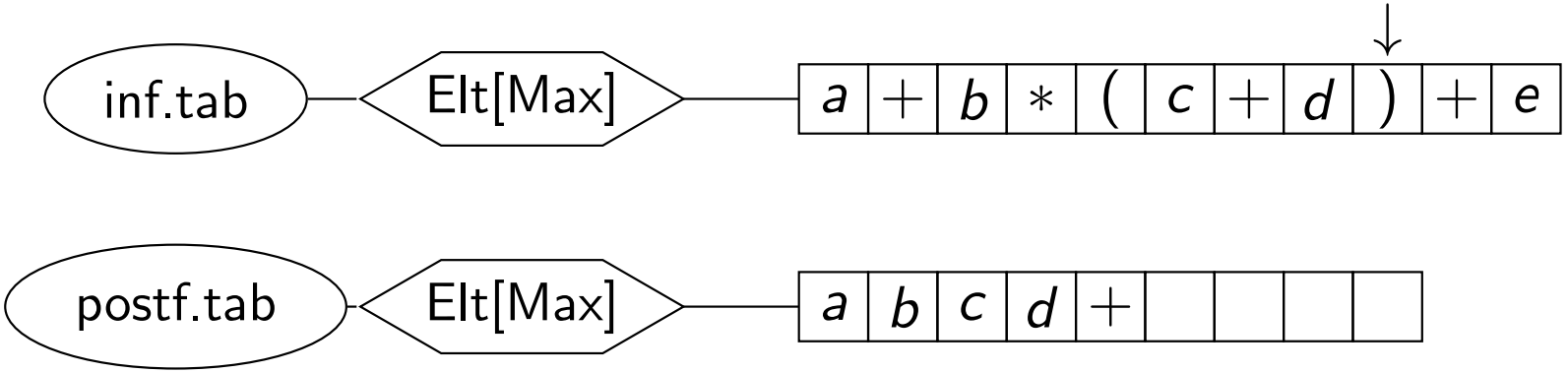
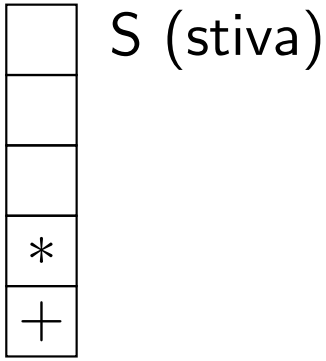
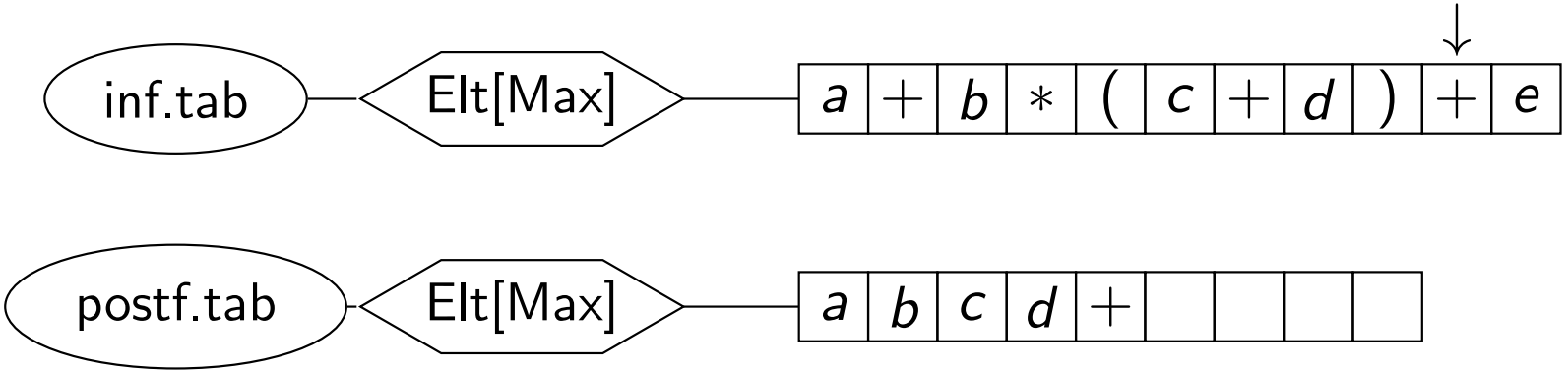
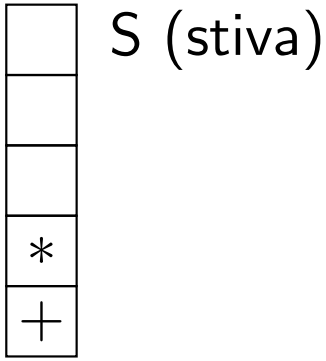




Conversia infixat → postixat

Exemplu: $a + b * (c + d) + e$





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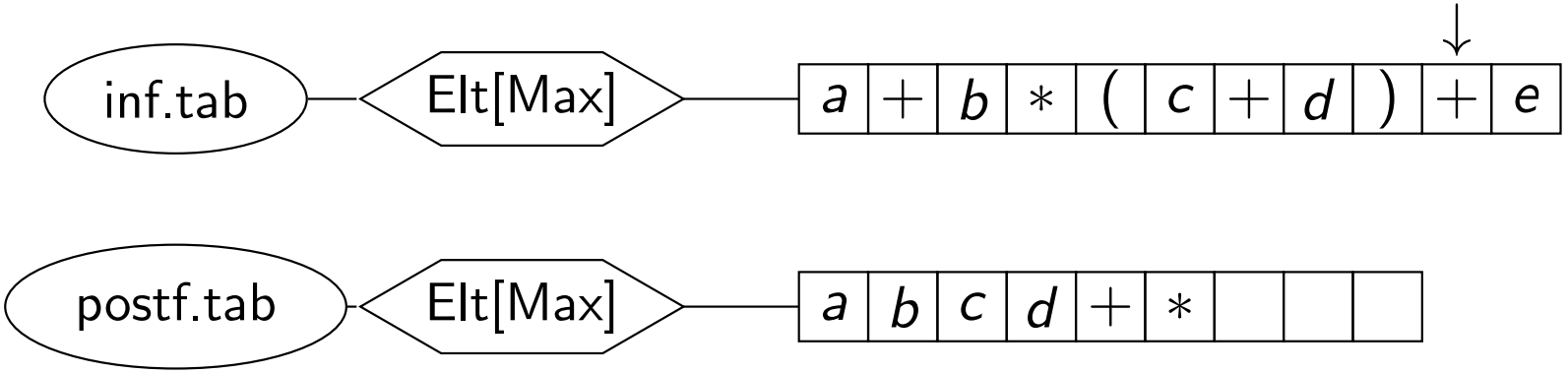

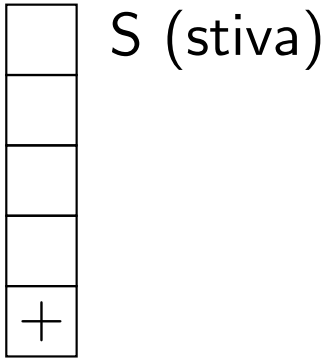

Conversia infixat → postixat

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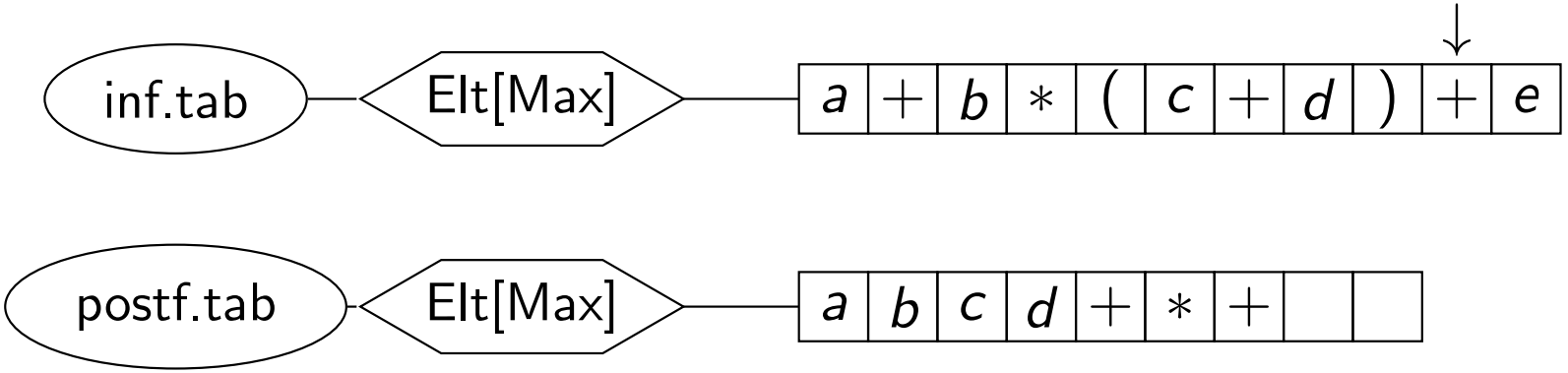
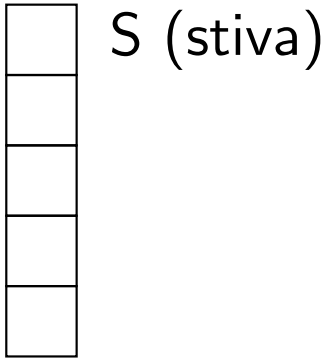
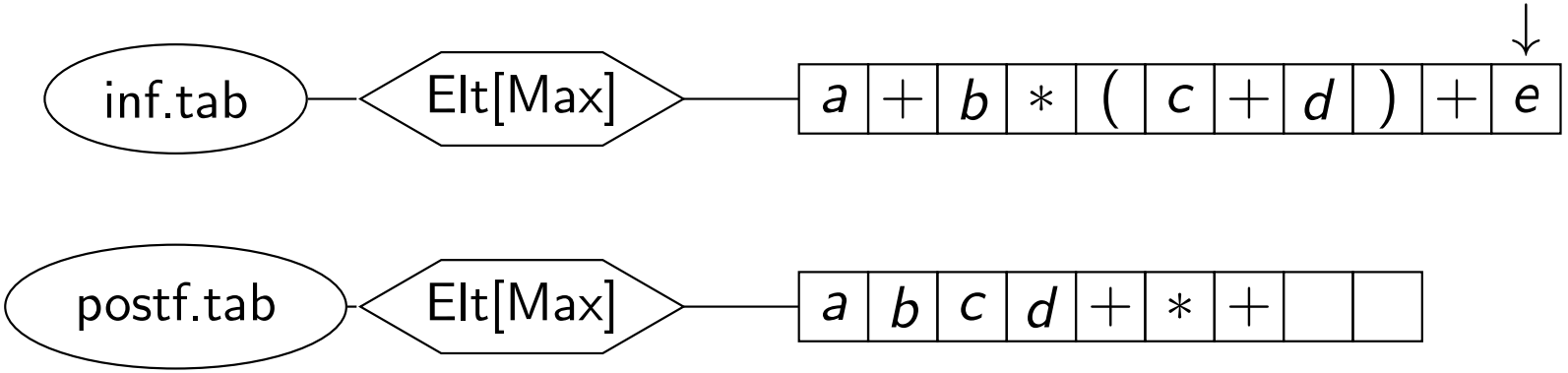
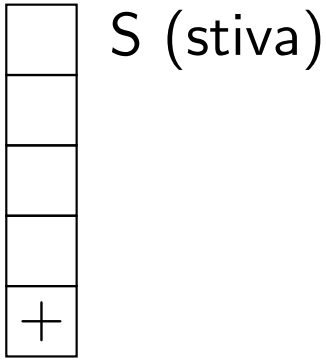




Conversia infixat → postixat

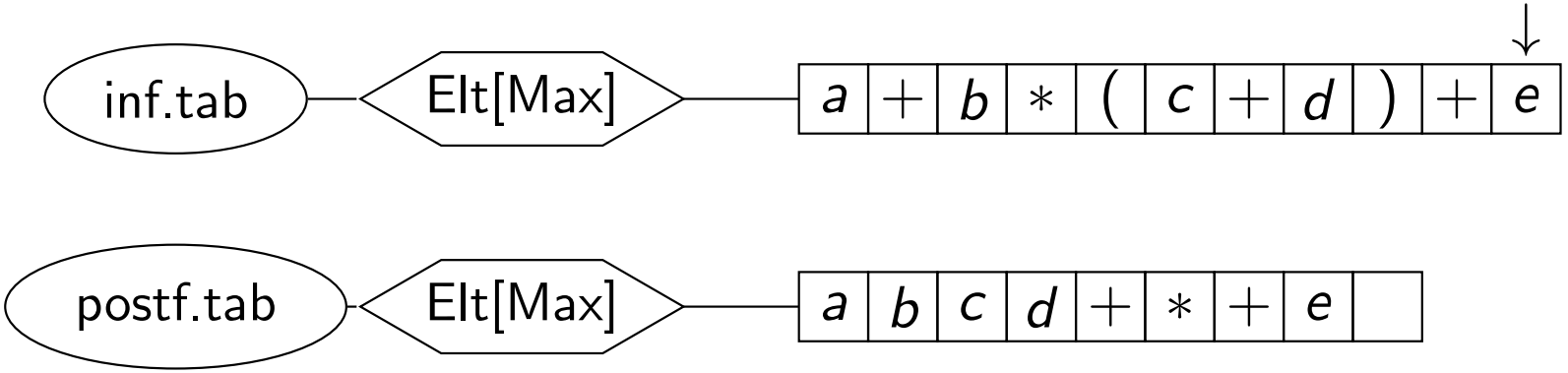
Exemplu: $a + b * (c + d) + e$

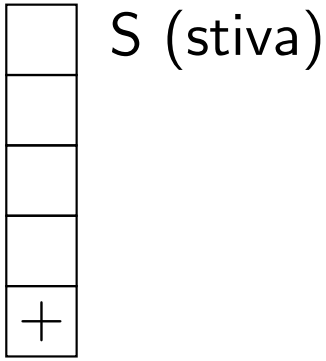




Conversia infixat → postixat

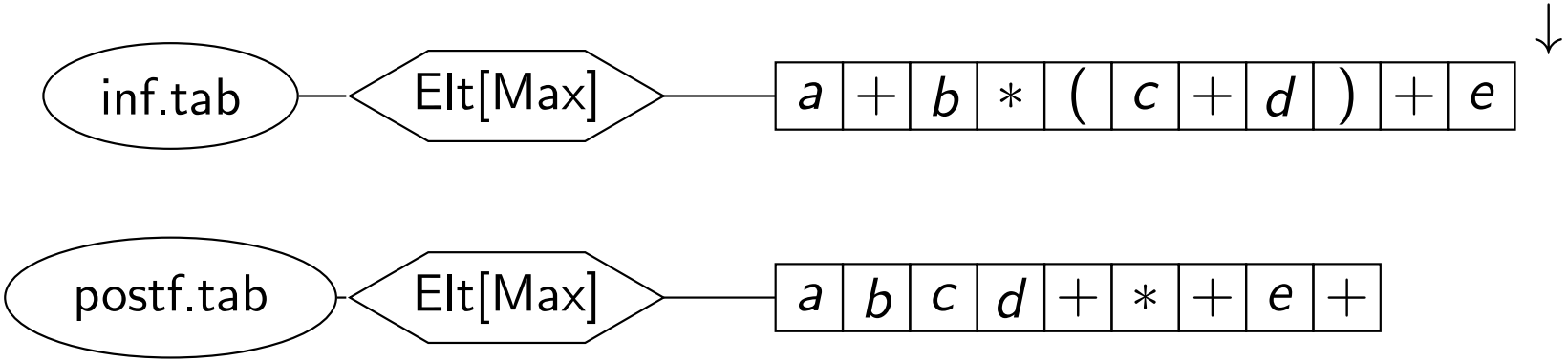
Exemplu: $a + b * (c + d) + e$

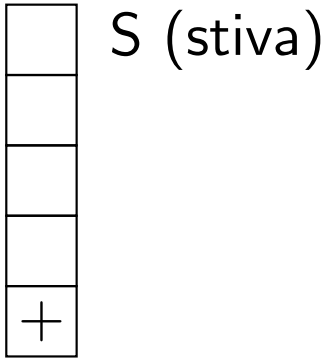




Conversia infixat → postixat

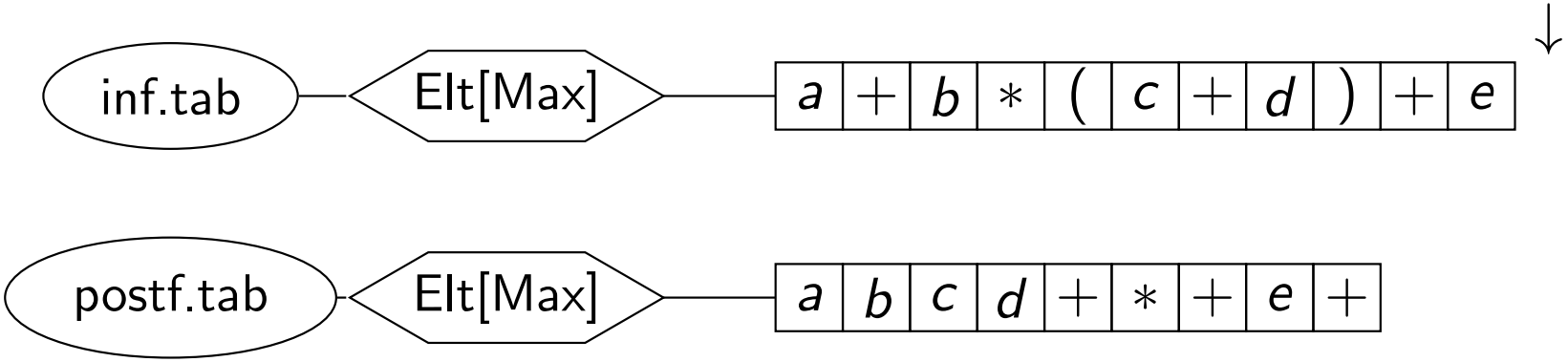
Exemplu: $a + b * (c + d) + e$

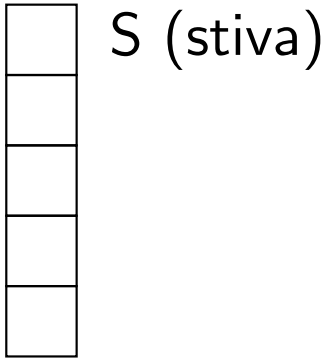




Conversia infixat → postixat

Exemplu: $a + b * (c + d) + e$



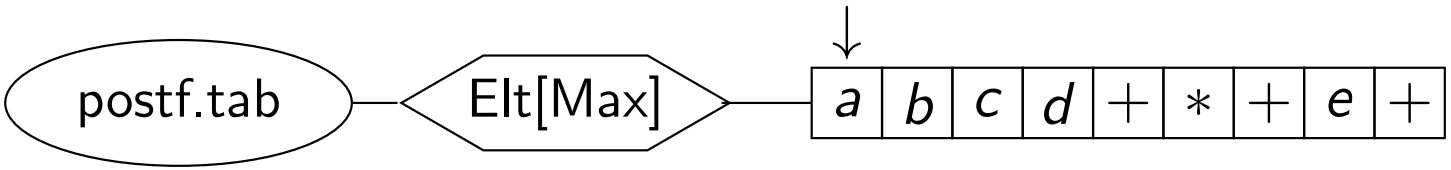


Conversia infixat → postfixat

```
procedure convInfix2Postfix(infix, postfix)
/* infix și postfix sunt cozi*/
begin
    S ← stivaVida()
    while (not esteVida(infix)) do
        x ← citeste(infix);    elimina(infix)
        if (operand(x) then
            insereaza(postfix, x)
        else
            if (x ==')') then
                while (top(S)! =' (') do
                    insereaza(postfix, top(S));    pop(S)
                pop(S)
            else
                while (not estevida(S) and top(S)! =' (' and
                  priorit(top(S)) >= priorit(x)) do
                    insereaza(postfix, top(S));    pop(S)
                push(S, x)
    while (not estevida(S)) do
        insereaza(postfix, top(S));    pop(S)
end
```

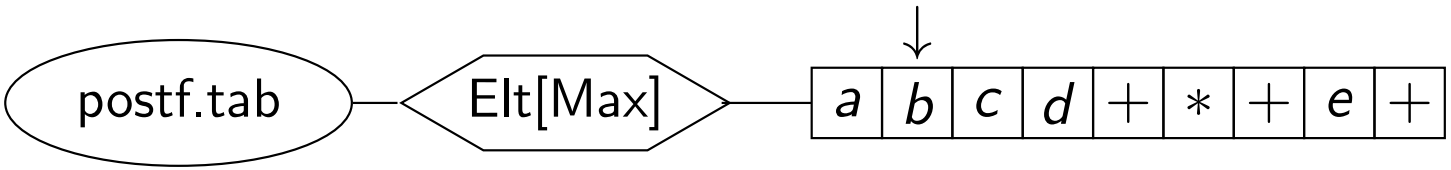
Evaluarea expresiilor postixate

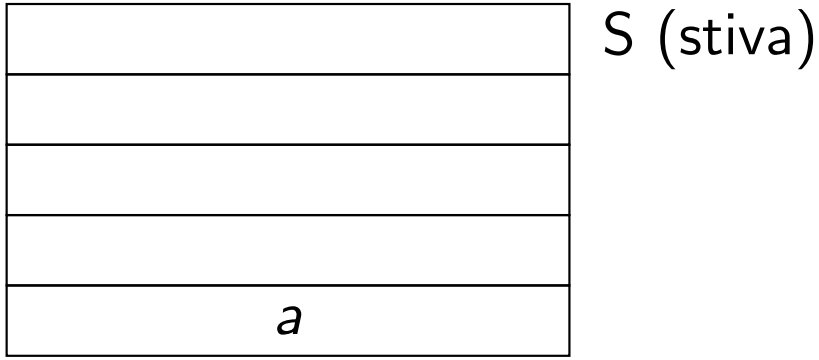
Exemplu: $a\ b\ c\ d\ +*+e+$



Evaluarea expresiilor postixate

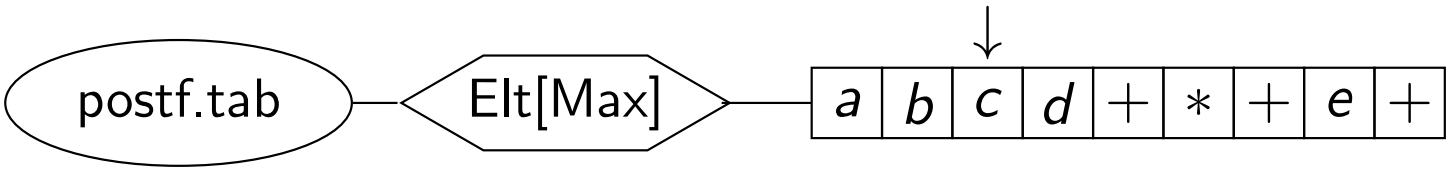
Exemplu: $a\ b\ c\ d\ +*+e+$

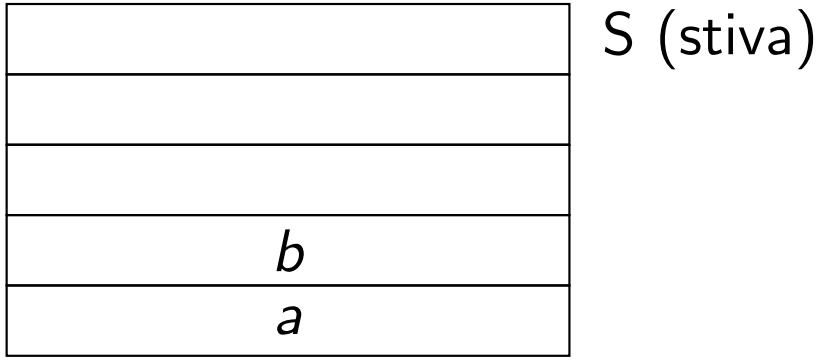




Evaluarea expresiilor postixate

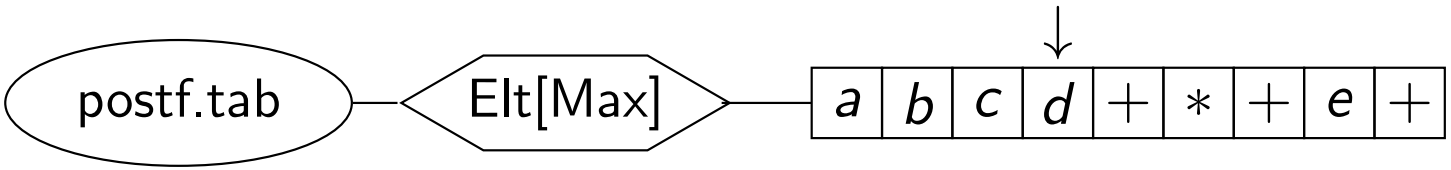
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

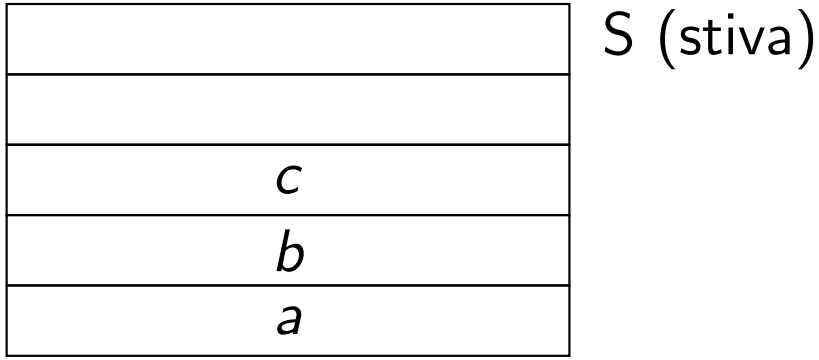




Evaluarea expresiilor postixate

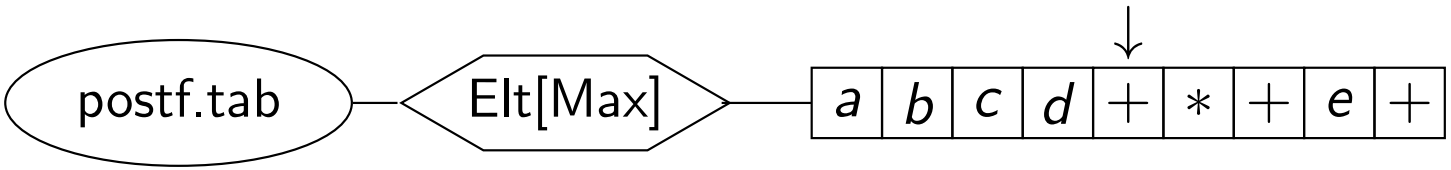
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

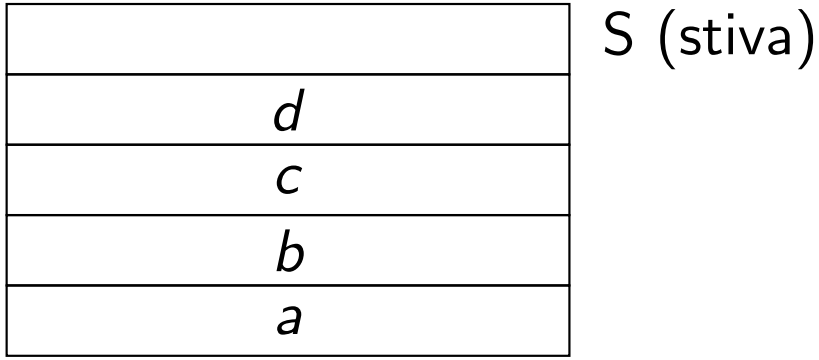




Evaluarea expresiilor postixate

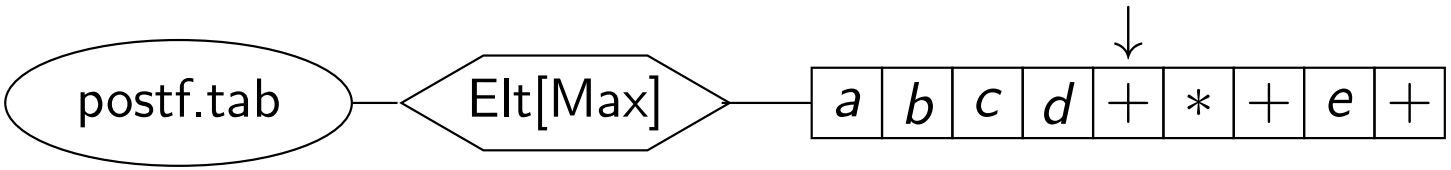
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

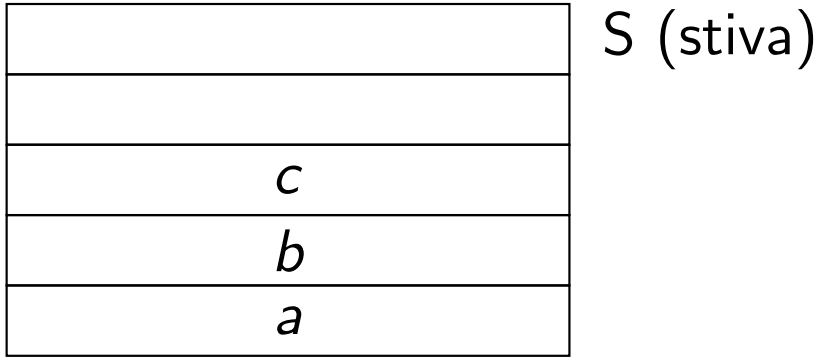




Evaluarea expresiilor postixate

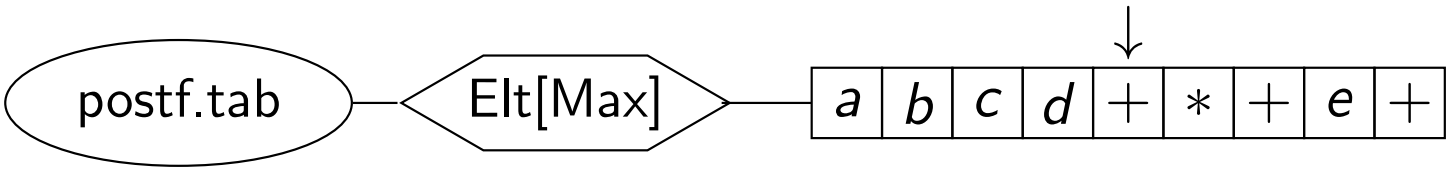
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

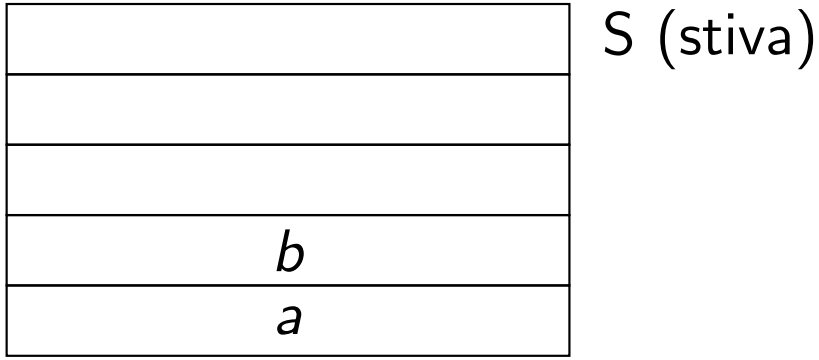




Evaluarea expresiilor postixate

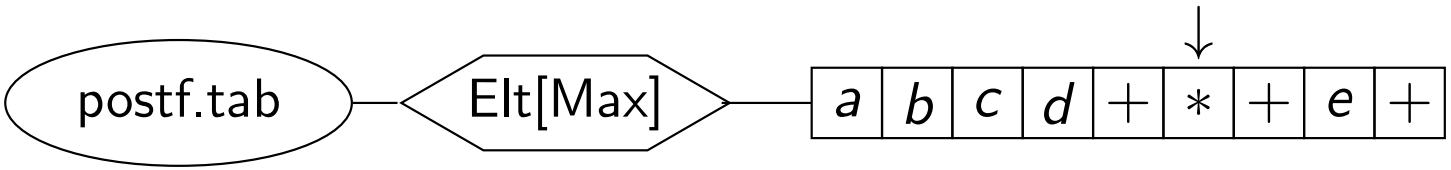
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

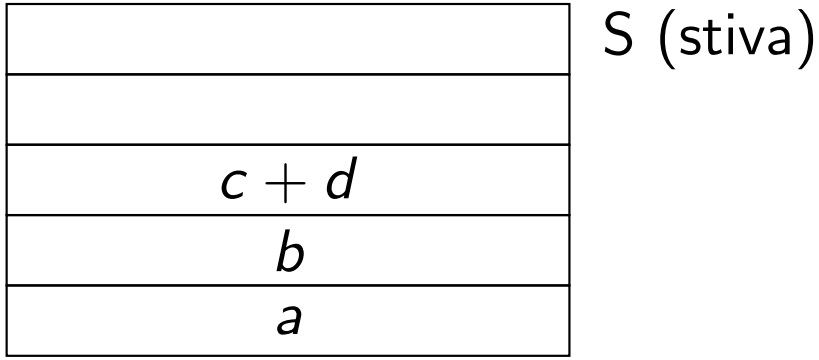




Evaluarea expresiilor postixate

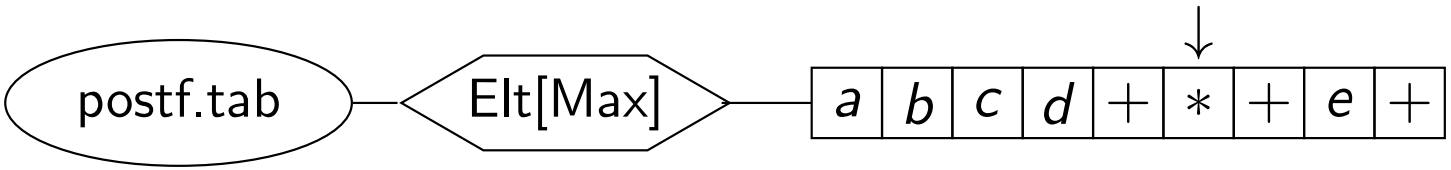
Exemplu: $a\ b\ c\ d\ +\ *\ +\ e\ +$

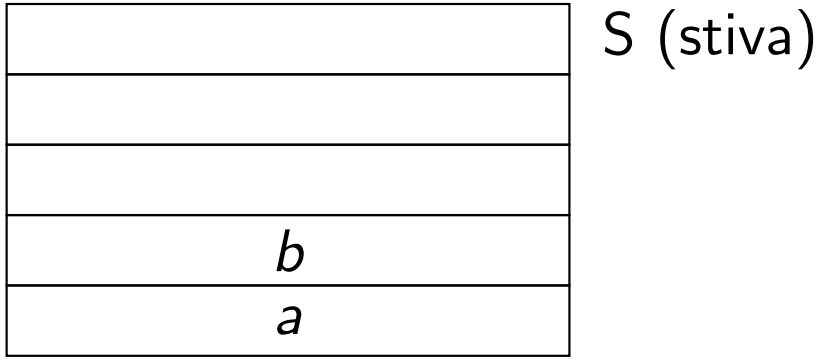




Evaluarea expresiilor postixate

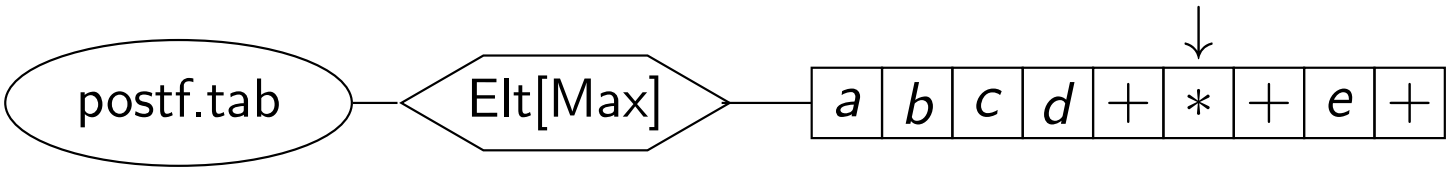
Exemplu: $a\ b\ c\ d\ +*+e+$

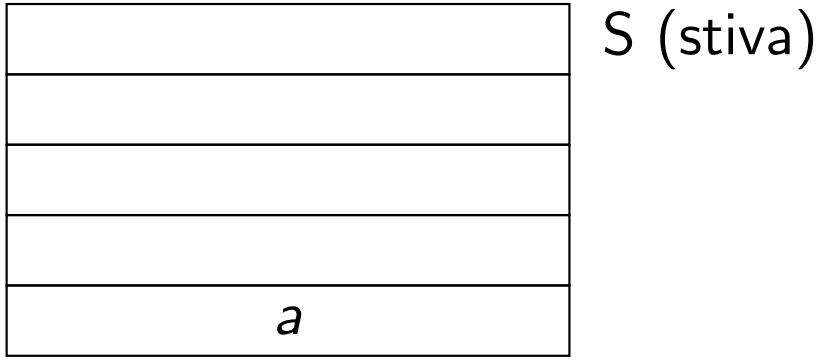




Evaluarea expresiilor postixate

Exemplu: $a\ b\ c\ d\ +*+e+$





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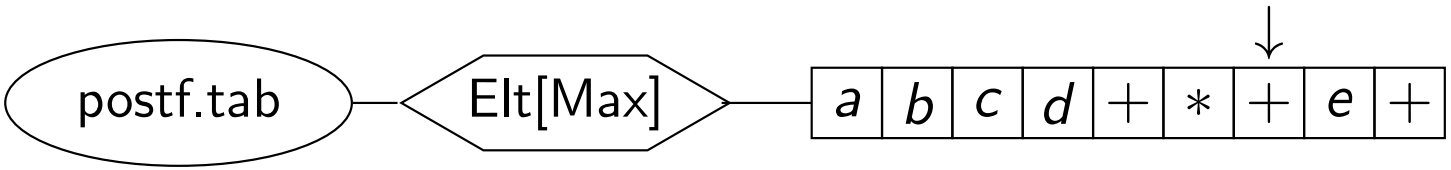
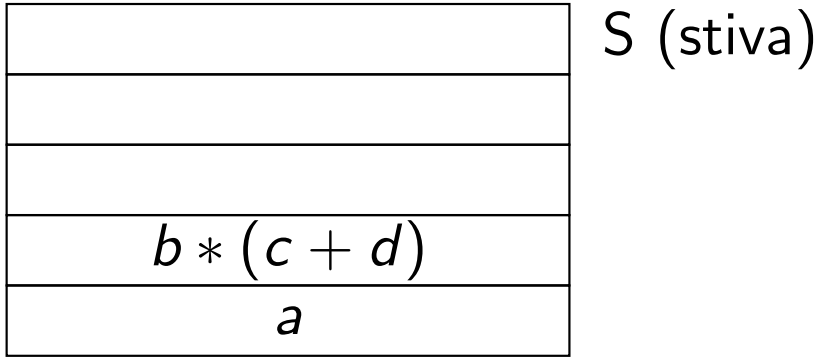
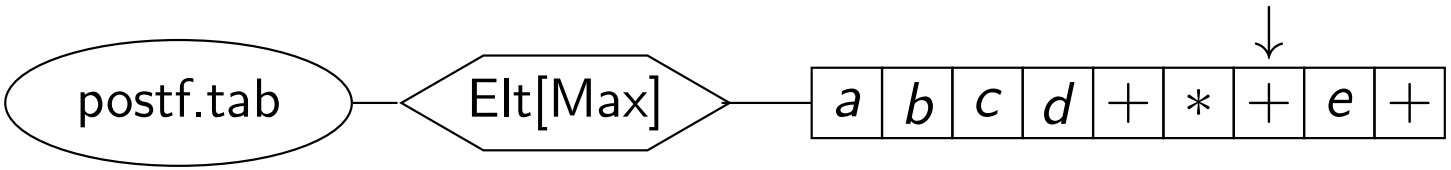
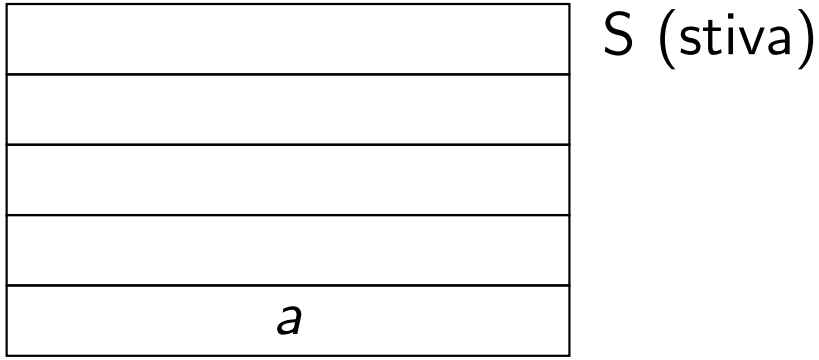




Evaluarea expresiilor postixate

Exemplu: $a\ b\ c\ d\ +*+e+$





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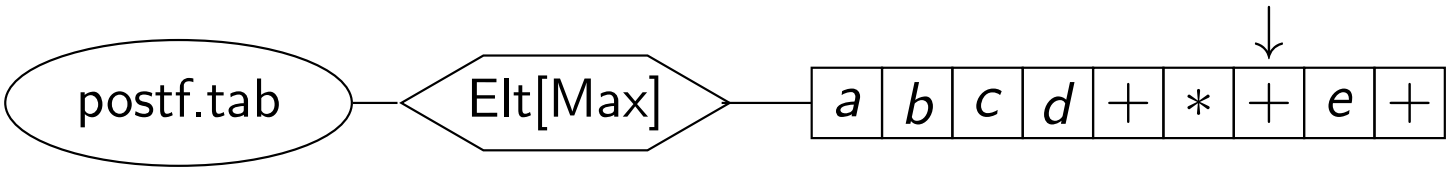

Evaluarea expresiilor postixate

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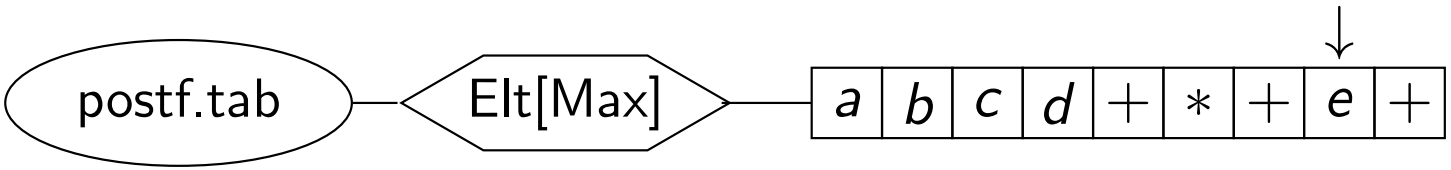
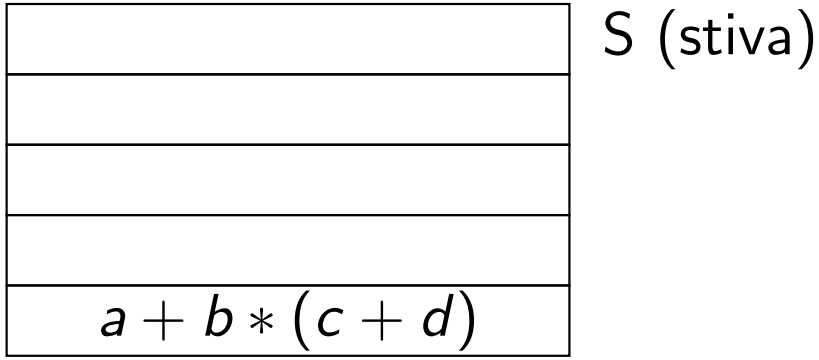
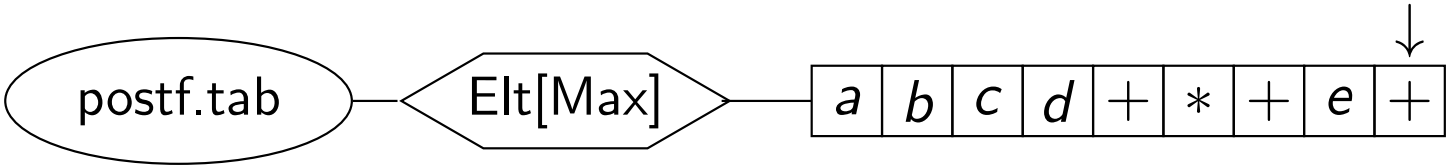
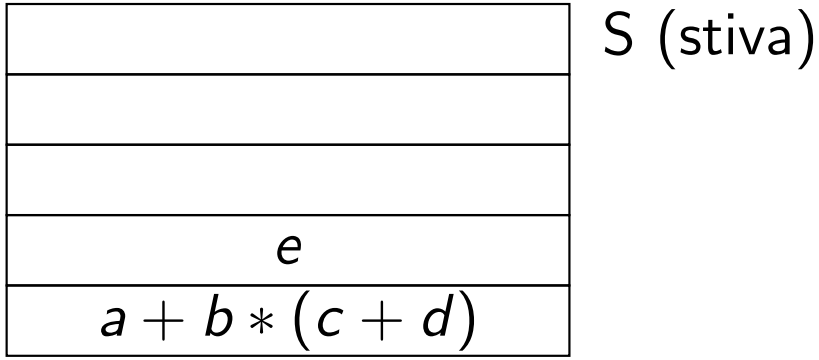




Evaluarea expresiilor postixate

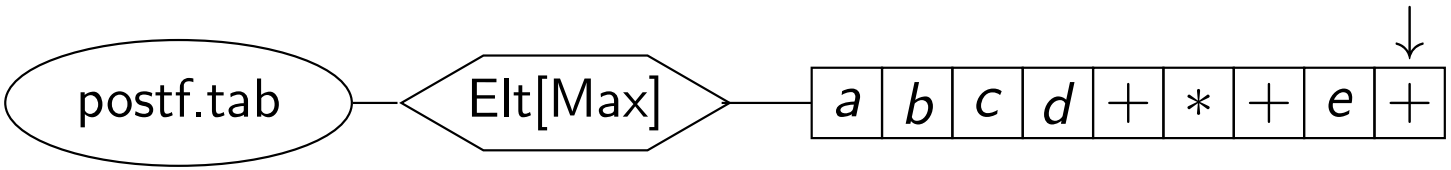
Exemplu: $a\ b\ c\ d\ +*+e+$

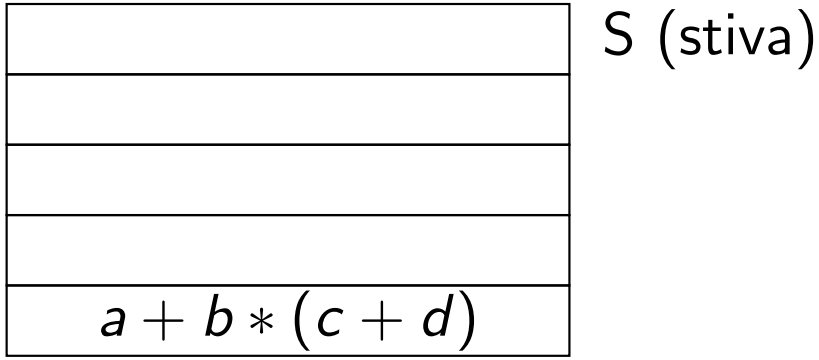




Evaluarea expresiilor postixate

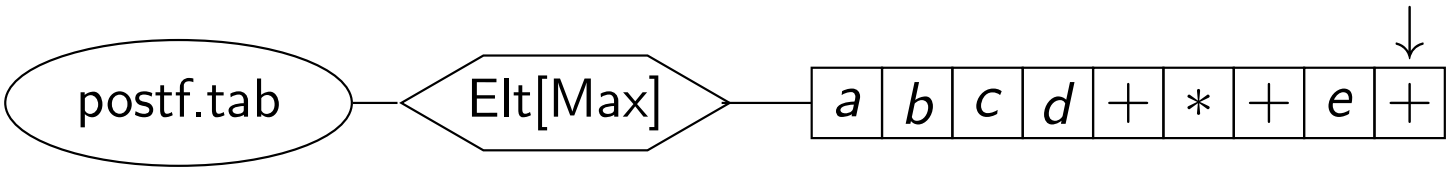
Exemplu: $a\ b\ c\ d\ +*+e+$

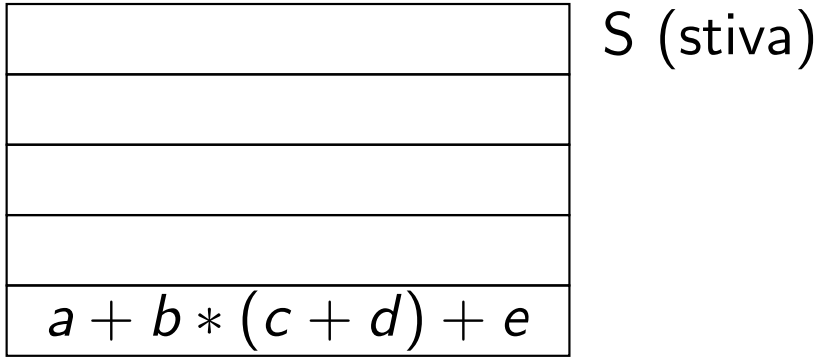




Evaluarea expresiilor postixate

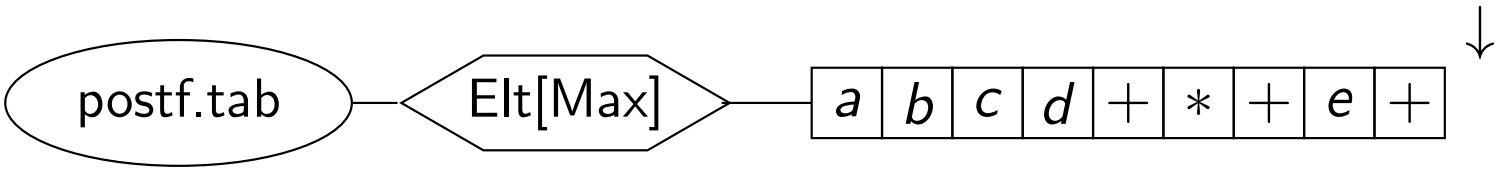
Exemplu: $a\ b\ c\ d\ +*+e+$

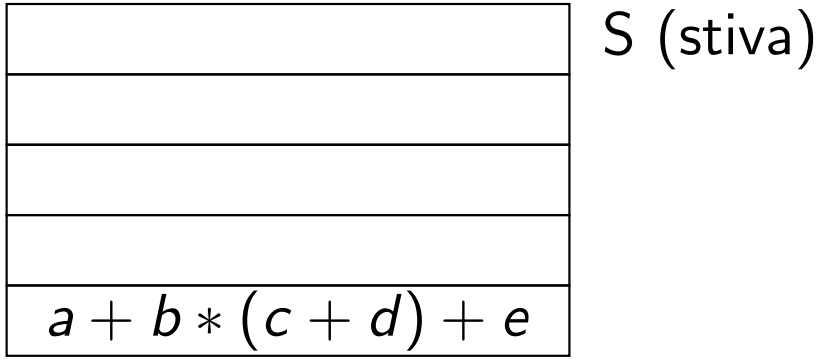




Evaluarea expresiilor postixate

Exemplu: $a\ b\ c\ d\ +*+e+$





Evaluarea expresiilor postfixate

```
function valPostfix(postfix)
begin
    S ← stivaVida()
    while (not esteVida(postfix)) do
        x ← citeste(postfix);    elimina(infix)
        if (operand(x) then
            push(S, x)
        else
            drp ← top(S);    pop(S)
            stg ← top(S);    pop(S)
            val ← stg op(x) drp
            push(S, val)
    val = top(S);    pop(S)
    return val
end
```